

The Intention Experiment (full text) by Lynne McTaggart. In HTML for free access. Part 2 of 4.

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This is part 2 of 4.

This is a complete reprint of the book titled “The Intention Experiment” by Lynne McTaggart. It is a non-fiction book, and it is groundbreaking. In this book, the author has compiled all those studies about the reality of ESP, and PSI, and compiled the results. The results are pretty damning. Something is going on, and Newtonian physics cannot explain it. It can only be explained with quantum physics.

What is going on is that quantum physics is working and weaving it’s magic throughout our lives, and rather than discount things as “superstition” and out-dated religion, this book connects actual scientific studies with the quantum physics principles involved. It explains so many thing that have been discounted as pure superstition.

Thus it’s placement in my blog.

This is for those people who want nice and clean answers to what is going on, yet cannot shake off the Newtonian physics that they learned in High School. This book

teaches you that there is a deeper reality behind everything and as such, it helps explain some elements of paranormal and religion that are often discounted as primitive nonsense.

Welcome to the world of quantum physics and how all those things about prayer, intention, and spirituality actually does have a scientific foundation that they are based upon.

CHAPTER THREE

The Two-Way Street

CLEVE BACKSTER WAS AMONG THE FIRST to propose that plants affected by human intention – a notion considered so preposterous that it was ridiculed for 40 years. Backster achieved his notoriety from a series of experiments that purported to demonstrate that living organisms read and respond to a person's thoughts.

Plant telepathy interested me less than a tangential discovery of his that has been sidelined amid all his adverse publicity: evidence of a constant two-way flow of information between all living things. Every organism, from bacteria to human beings, appears to be in perpetual quantum communication. This relentless conversation offers a ready mechanism by which thoughts can have a physical effect.

This discovery resulted from a silly little diversion in 1966; Backster, at the time a tall, wiry man with a buzz cut and a great deal of childlike enthusiasm, was easily distracted. He often carried on working in his suite of offices when the rest of his staff had gone home and he could finally focus without the constant interruptions of colleagues and the tumultuous daytime activity of Times Square, four storeys below.¹

Backster had made his name as the country's leading lie-detector expert. During the Second World War, he had been fascinated by the psychology of lying, and the use of hypnosis and 'truth serum' interrogation in counter-intelligence, and he had brought these twin fascinations to bear in refining the polygraph test to a high psychological art. He had launched his first programme with the CIA for counter-intelligence several years after the war, and then went on to found the Backster School of Lie Detection, still the world's leading school teaching polygraph techniques some 50 years after it first opened its doors.

One morning in February, after working all night, Backster was taking a coffee break at 7 a.m. He was about to water the *Dracaena* and rubber plant in his office. As he filled up his watering can, he wondered if it might be possible to measure the length of time it would take water to travel up the stem of a plant from the roots and reach the leaves, particularly in the *Dracaena*, a cane plant with an especially long trunk. It

occurred to him that he could test this by connecting the *Dracaena* to one of his polygraph machines; once the water reached the spot between the electrodes, the moisture would contaminate the circuit and be recorded as a drop in resistance.

A lie detector is sensitive to the slightest change in the electrical conductivity of skin, which is caused by increased activity of the sweat glands, which in turn are governed by the sympathetic nervous system. The polygraph galvanic skin response (GSR) portion of the test displays the amount of the skin's electrical resistance, much as an electrician's ohmmeter records the electrical resistance of a circuit. A lie detector also monitors changes in blood pressure, respiration, and the strength and rate of the pulse. Low levels of electrical conductivity indicate little stress and a

state of calm. Higher electrodermal activity (EDA) readings indicate that the sympathetic nervous system, which is sensitive to stress or certain emotional states, is in overdrive – as would be the case when someone is lying. A polygraph reading can offer evidence of stress to the sympathetic nervous system even before the person being tested is consciously aware of it.

In 1966, the state-of-the-art technology consisted of a set of electrode plates, which were attached to two of a subject's fingers, and through which a tiny current of electricity was passed. The smallest increases or decreases in electrical resistance were picked up by the plates and recorded on a paper chart, on which a pen traced a continuous, serrated line. When someone lied or in any way experienced a surge of emotion (such as excitement or fear), the size of the zigzag would dramatically increase and the tracing would move to the top of the chart.

Backster sandwiched one of the long, curved leaves of the *Dracaena* between the two sensor electrodes of a lie detector and encircled it with a rubber band. Once he watered the plant, what he expected to see was an upward trend in the ink tracing on the polygraph recording paper, corresponding to a drop in the leaf's electrical resistance as the moisture content increased. But as he poured in the water, the very opposite occurred. The first part of the tracing began heading downward and then displayed a short-term blip, similar to what happens when a person briefly experiences a fear of detection.

At the time Backster thought he was witnessing a human-style reaction, although he would later learn that the waxy insulation between the cells in plants causes an electrical discharge that mimics a human stress reaction on polygraph instruments. He decided that if the plant were indeed displaying an emotional reaction, he would have to come up with some major emotional stimulus to heighten this response.

When a person takes a polygraph test, the best way to determine if he is lying is to ask a direct and pointed question, so that any answer but the truth will cause an

immediate, dramatic stress reaction in his sympathetic nervous system: 'Was it you who fired the two bullets into Joe Smith?'

In order to elicit the equivalent of alarm in a plant, Backster knew he needed somehow to threaten its well-being. He tried immersing one of the plant's leaves in a cup of coffee, but that did not cause any interesting reaction on the tracing – only a continuation of the downward trend. If this were the tracing of a human being, Backster would have concluded that the person being monitored was tired or bored. It was obvious to him that he needed to pose an immediate and genuine threat: he would get a match and burn the electroded leaf.

At the very moment he had that thought, the recording pen swung to the top of the polygraph chart and nearly jumped off. He had not burned the plant; he had only *thought* about doing so. According to his polygraph, the plant had perceived the thought as a direct threat and registered extreme alarm. He ran to his secretary's desk in a neighbouring office for some matches. When he returned, the plant was still registering alarm on the polygraph. He lit a match and flickered it under one of the leaves. The pen continued on its wild, zigzag course. Backster then returned the matches to his secretary's desk. The tracing calmed down and began to flat-line.

He hadn't known what to make of it. He had long been drawn to hypnosis and ideas about the power of thought and the nature of consciousness. He had even performed a number of experiments with hypnosis during his work with the Army Counter Intelligence Corps and the CIA, as part of a campaign designed to detect the use of hypnosis techniques in Russian espionage.

But this was something altogether more extraordinary. This plant, it seemed, had *read his thoughts*. It wasn't even as though he particularly *liked* plants. This only could have occurred if the plant possessed some sort of sophisticated extrasensory perception. The plant somehow must be attuned to its environment, able to receive far more than pure sensory information from water or light.

Backster modified his polygraph equipment to amplify electrical signals so that they would be highly sensitive to the slightest electrical change in the plants. He and his partner, Bob Henson, set about replicating the initial experiment. Backster spent the next year and a half frequently monitoring the reactions of the other plants in the office to their environment. They discovered a number of characteristics. The plants grew attuned to the comings and goings of their main caretaker. They also maintained some sort of 'territoriality' and so did not react to events in the other offices near Backster's lab. They even seemed to tune in to Pete, his Doberman Pinscher, who spent his days at the office.

Most intriguing of all, there seemed to be a continuous two-way flow of information between the plants and other living things in their environment. One day, when Backster boiled his kettle to make coffee, he found he had put in too much water. But when he poured the residue down the sink, he noticed that the plants registered an intense reaction.

The sink was not the most hygienic; indeed, his staff had not cleaned the drain for several months. He decided to take some samples from the drain and examine them under a microscope, which showed a jungle of bacteria that ordinarily lives in the waste pipes of a sink. When threatened by the boiling water, had the bacteria emitted a type of mayday signal before they died, which had been picked up by the plants?

Backster, who knew he would be ridiculed if he presented findings like these to the scientific community, enlisted an impressive array of chemists, biologists, psychiatrists, psychologists and physicists to help him design an airtight experiment. In his early experiments, Backster had relied upon human thought and emotion as the trigger for reactions in the plants. The scientists discouraged him from using intention as the stimulus of the experiment, because it did not lend itself to rigorous scientific design. How could you set up a control for a human thought – an intention to harm, say? The orthodox scientific community could easily pick holes in his study. He had to create a laboratory barren of any other living things besides the plants to ensure that the plants would not be, as it were, distracted.

The only way to achieve this was to automate the experiment entirely. But he also needed a potent stimulus. He tried to think of the one act that would stir up the most profound reaction, something that would evoke the equivalent in the plants of dumbfounded horror. It became clear that the only way to get unequivocal results was to commit the equivalent of mass genocide. But what could he kill *en masse* that would not arouse the ire of anti-vivisectionists or get him arrested? It obviously could not be a person or a large animal of any variety. He did not even want to kill members of the usual experimental population, like rats or guinea pigs. The one obvious candidate was brine shrimp. Their only purpose, as far as he could tell, was to become fodder for tropical fish. Brine shrimp were already destined for the slaughterhouse. Only the most ardent anti-vivisectionist could object.

Backster and Henson rigged up a gadget that would randomly select one of six possible moments when a small cup containing the brine shrimp would invert and tip its contents into a pot of continuously boiling water. The randomizer was placed in the far room in his suite of six offices, with three plants attached to polygraph equipment in three separate rooms at the other end of the laboratory. His fourth polygraph machine, attached to a fixed valve resistor to ensure that there was no sudden surge of voltage from the equipment, acted as the control.

Microcomputers had yet to be invented, as Backster set up his lab in the late sixties. To perform the task, Backster created an innovative mechanical programmer, which operated on a time-delay switch, to set off each event in the automation process. After flipping the switch, Backster and Henson would leave the lab, so they and their thoughts would not influence the results. He had to eliminate the possibility that the plants might be more attuned to him and his colleague than a minor murder of brine shrimp down the hallway.

Backster and Henson tried their test numerous times. The results were unambiguous: the polygraphs of the electroded plants spiked a significant number of times just at the point when the brine shrimp hit the boiling water.

Years after he had made this discovery – and after he became a great fan of *Star Wars* – he would think of this moment as one in which his plants picked up a major disturbance in the Force, and he had discovered a means of measuring it.[2](#)

If plants could register the death of an organism three doors away, it must mean that all life forms were exquisitely in tune with each other. Living things must be registering and passing telepathic information back and forth at every moment, particularly at moments of threat or death.

Backster published the results of his experiment in several respected journals of psychic research and gave a modest presentation before the Parapsychology Association during its tenth annual meeting.[3](#)

Parapsychologists recognized Backster's contribution and replicated it in a number of independent laboratories, notably that of Alexander Dubrov, a Russian doctor of botany and plant physiology.[4](#) It was even glorified in a bestselling book, *The Secret Life of Plants*.[5](#) But among the mainstream scientific community, his research was disparaged as ludicrous, largely because he was not a traditional scientist, and he was ridiculed for what became known as 'The Backster Effect'. In 1975, *Esquire* magazine even awarded him one of its 100 Dubious Achievement Awards: 'Scientist claims yogurt talks to itself'.[6](#)

Nonetheless, over the next 30 years Backster ignored his critics and stubbornly carried on with his research, as well as his polygraph business, eventually amassing file drawers full of studies of what he referred to as 'primary perception'. A variety of plants that had been hooked up to his polygraph equipment showed evidence of a reaction to human emotional highs and lows, especially threats and other forms of negative intention – as did paramecia, mould cultures, eggs and, indeed, yogurt.[7](#)

Backster even demonstrated that bodily fluids such as blood and semen samples taken from himself and his colleagues registered reactions mirroring the emotional

state of their hosts; the blood cells of a young lab assistant reacted intensely the moment he opened a *Playboy* centrefold and caught sight of Bo Derek in the nude.[8](#)

These reactions were not dependent on distance; any living system attached to a polygraph reacted similarly to his thoughts, whether he was in the room or miles away. Like pets, they had become attuned to their 'owner'.

These organisms were not simply registering his thoughts; they were communicating telepathically with all the living things in their environment. The live bacteria in yogurt displayed a reaction to the death of other types of bacteria and even evidenced a desire to be 'fed' with more of its own beneficial bacteria. Eggs registered a cry of alarm and then resignation when one of their number was dropped in boiling water. Plants appeared to react in real time to any break in continuity with the living beings in their environment. They even



Bo Derek

appeared to react at the moment when their caretakers, who were away from the office, decided to return.[9](#)

His major difficulty was designing experiments that could demonstrate an effect scientifically. Even though his laboratory experiments were now entirely automated, when he left the office, the plants would remain attuned to him, no matter now far away he went. If Backster and his partner were at a bar a block away during an experiment, he would discover that the plants were not responding to the brine shrimp, but to the rising and falling animation of their conversations. It got so difficult to isolate reactions to specific events that eventually he had to design experiments that would be carried out by strangers in another lab.

Repeatability remained another big problem. Any tests required spontaneity and true intent. He had discovered this when the famous remote viewer Ingo Swann had come to visit him at his lab in October 1971.

Swann wanted to repeat Backster's initial experiment with his *Dracaena*. As expected, the plant's polygraph began to spike when Swann imagined burning the plant with a match. He tried it again, and the plant reacted wildly, then stopped.

'What does *that* mean?' Swann asked. Backster shrugged. 'You tell me.'

The thought that occurred to Swann was so bizarre that he was not sure whether to say it aloud. 'Do you mean,' he said, 'that it has *learned* that I'm not serious about really burning its leaf? So that it now knows it need not be alarmed?'

'*You* said it, I didn't,' Backster replied. 'Try another kind of harmful thought.'

Swann thought of putting acid in the plant's pot. The needle on the polygraph again began to zigzag wildly. Eventually, the plant appeared to understand that Swann was not serious. The polygraph tracing flat-lined. Swann, a plant lover who was already convinced that plants were sentient, was nevertheless shocked at the thought that plants could learn to differentiate between true and artificial human intent: *a plant learning curve*.[10](#)

Although certain questions remain about Backster's unorthodox research methods, the sheer bulk of his evidence argues strongly for some sort of primary responsiveness and attuning, if not sentience, present in all organisms, no matter how primitive. But for my purposes, Backster's real contribution was his discovery of the telepathic communication carrying on between every living thing and its environment. Somehow, a constant stream of messages was being sent out, received and replied to.

Backster had to wait some years to discover the mechanism of this communication, which became apparent when physicist Fritz-Albert Popp discovered biophotons.[11](#)

At first Popp believed that a living organism used biophoton emissions solely as a means of instantaneous, non-local signalling from one part of the body to another – to send information about the global state of the body's health, say, or the effects of any particular treatment. But then Popp grew intrigued by the most fascinating effect of all: the light seemed to be a communications system *between* living things.[12](#)

In experiments with *Daphnia*, a common water flea, he discovered that female water fleas were absorbing the light emitted from each other and sending back wave interference patterns, as though they had taken the light sent to themselves and updated it with more information. Popp concluded that this activity may be the

mechanism enabling fleas to stay together when they swarm – a silent communication holding them together like an invisible net.[13](#)

He decided to examine the light emissions between dinoflagellates, luminescent algae that cause phosphorescence in seawater. These single-celled organisms sit somewhere between an animal and a plant in the evolutionary scale; although they are classified as a plant, they move like a primitive animal. Popp discovered that the light of each dinoflagellate was coordinated with that of its neighbours, as if each were holding aloft a tiny lantern on cue.[14](#)

Chinese colleagues of Popp's who had tried positioning two samples of the algae so that they could 'see' each other through a shutter also found that the light emissions from each sample were synchronous. The researchers concluded that they had witnessed a highly sophisticated means of communication. There was no doubt that the two samples were signalling to each other.[15](#)

These organisms also appeared to be registering light from other species, although the greatest synchronicities occurred between members of the same species.[16](#)

Once the light waves of one organism were initially absorbed by another organism, the first organism's light would begin trading information in synchrony. [17](#)

Living things also appeared to communicate information with their surroundings. Bacteria absorbed light from their nutritional media: the more bacteria present, Popp found, the greater the absorption of light.[18](#)

Even the white and yolk of an egg appear to communicate with the shell.[19](#)

This communication carries on, even if an organism is cut into pieces. Gary Schwartz cut up a batch of string beans, placed them between 1 millimetre and 10 millimetres apart, and then used the NSF CCD camera he had borrowed to take series of photographs of the sections. Using software to enhance the light between the beans, he discovered so much light between the sections that it appeared as though the bean were whole again. Even though the string beans had been severed, the individual sections carried on their communication to the rest of the vegetable.[20](#)

This may be the mechanism accounting for the feeling described by amputees with phantom limb sensations. The light of the body still communicates with the energetic 'footprint' of the amputated limb.

Like Backster, Popp discovered that living things are exquisitely in tune with their environment through these light emissions. One of Popp's colleagues, Professor Wolfgang Klimek, the head of the Ministry of Research for the German government devised an ingenious experiment to examine whether creatures such as algae were

aware of past disturbances in their environment. He prepared two containers of seawater, and shook one of them. After 10 minutes, when the water in the shaken container had settled down, he placed samples of dinoflagellates in the two vessels. Those algae exposed to the shaken water suddenly increased their photon emissions – a sign of stress. The algae appeared to be aware of the slightest change in their environment – even a historical change – and responded with alarm.[21](#)

Another of Popp's colleagues, Eduard Van Wijk, a Dutch psychologist, wondered how far this influence extended. Did a living thing register information from the entire environment, and not simply between two communicating entities? When a healer sends out healing intention, for instance, how far does his field of influence extend? Would he only affect his target, or would his aim have a shotgun effect, affecting other living organisms around the target?

Van Wijk placed a jar of *Acetabularia acetabulum*, another simple algae, near a healer and his patient, then measured the photon emissions of the algae during healing sessions and periods of rest. After analysing the data, he discovered remarkable alterations in the photon count of the algae. The quality of emissions significantly changed during the healing sessions, as though the algae were being bombarded with light. There also seemed to be changes in the rhythm of the emissions, as though the algae had become attuned to a stronger source of light.

During his initial research, Popp had discovered a strange reaction to light by a living thing. If he shone a bright light on an organism, after a certain delay, the organism would shine more brightly itself with extra photons, as if it were rejecting any excess. Popp called this phenomenon 'delayed luminescence', and assumed it was a corrective device to help the organism maintain its level of light at a delicate equilibrium. In Van Wijk's experiment, the photon emissions of algae showed highly significant shifts from normal, when plotted on a graph. Van Wijk had generated some of the first evidence that healing light may affect anything in its path.[22](#)

Gary Schwartz's associate Melinda Connor then demonstrated that intention has a direct effect on this light. For her study she clipped leaves from geranium plants, carefully matching them in pairs for size, health, placement on the plant and access to light and close to identical photon emissions. She asked each of 20 master energy healers to send intentions to one of each pair of leaves, first to reduce emissions and then to increase them. In 29 of the 38 sessions designed to decrease emissions, the light was significantly lowered in the treatment leaves, and in 22 of the 38 trials intending to increase the light, the healers caused a significantly greater glow.[23](#)

Sometimes a physical jolt to the system triggers a shock of realization. For physicist Konstantin Korotkov, his insight resulted from a fall off a roof. It was the winter of

1976, and Korotkov, who was 24 at the time, had been celebrating a birthday with some friends. Korotkov liked to celebrate outside, whatever the weather. He and his friends had been drinking vodka on the roof. Korotkov was given to expansive gestures, and during a moment of gaiety, threw himself off the roof onto what he thought was a deep bed of snow, which he assumed would cushion his fall. But hidden beneath the snow lay hard stone. Korotkov broke his left leg and landed in the hospital for months.[24](#)

During his long recovery, Korotkov, a conventional professor of quantum physics at St Petersburg State Technical University in Russia, pondered on a lecture on Kirlian effects and healing that he had attended earlier that year. He had been so intrigued that he wondered if he could improve on what Kirlian claimed to be doing: capturing someone's life energy on film.

Semyon Davidovich Kirlian was an engineer who had discovered in 1939 that photographing living things that had been exposed to a pulsed electromagnetic field would capture what many have termed the human 'aura'. When any conductive object (like living tissue) is placed on a plate made of an insulating material, such as glass, and exposed to high-voltage, high-frequency electricity, a low current results that creates a corona discharge, a halo of coloured light around the object that can be captured on film. Kirlian claimed that the state of the aura reflected the person's state of health; changes in the aura were evidence of disease or mental disturbance.

The Soviet scientific mainstream ignored Kirlian until the 1960s, when the Russian press discovered bioelectrography, as it came to be called, and hailed him as a great inventor. Kirlian photography suddenly became respectable, particularly in space research, and was championed by many Western scientists. Publication of Kirlian's first study in 1964 further attracted the scientific community.[25](#)

Lying for months in his bed, Korotkov realized that if he was going to discover more about how to capture this mysterious light Kirlian claimed was so vital to health, he was going to have to give up his day job. He knew that the involvement of a well-established quantum physicist such as himself would lend the technique scientific legitimacy and his technical ability might also help advance the technology. Perhaps he could even devise a means of depicting the light in real time.

After he got back up on his feet, Korotkov spent months developing a mechanism, which he called the Gas Discharge Visualization (GDV) technique, that made use of state-of-the-art optics, digitized television matrices and a powerful computer. Ordinarily, a living thing will dribble out the faintest pulse of photons, perceptible only to the most sensitive equipment in conditions of utter pitch black. As Korotkov realized, a better way to capture this light was to stir up photons by 'evoking', or

stimulating them into an excited state so that they would shine millions of times more intensely than normal.

His equipment blended several techniques: photography, measurements of light intensity and computerized pattern recognition. Korotkov's camera would take pictures of the field around each of the 10 fingers, one finger at a time. A computer program would then extrapolate from this a real-time image of the 'biofield' surrounding the organism and deduce from it the state of the organism's health.

Korotkov went on to write five books on the human bioenergy field.[26](#)

In time, he managed to convince the Russian Ministry of Health of the importance of his invention to medical technology, diagnosis and treatment. His equipment was initially employed to predict certain clinical situations, such as the progress of recovery of people after surgery.[27](#)

It soon became widely used in Russia as a diagnostic tool for many illnesses, including cancer and stress,[28](#) and was even used to assess athletic potential – to predict the psychophysical reserves in athletes training for the Olympics and the likelihood of victory or exhaustion from overtraining.[29](#)

Eventually, some 3000 doctors, practitioners and researchers worldwide came to use the technology. The National Institutes of Health got interested and funded work on the 'biofield', which employed Korotkov's equipment.[30](#)

While officially exploring these practical applications, Korotkov privately carried on with his own studies of what had really captured his imagination: the connection between biofields and consciousness.[31](#)

He took GDV readings of healers and a *Qigong* master while they were sending energy, and discovered remarkable changes in their corona discharges. Korotkov then explored the effects of a person's thoughts on the people surrounding him. He asked a number of couples to 'send' a variety of thoughts to their partners, while they were standing within close range. Every strong emotion – whether love, hate or anger – produced an extraordinary effect on the light discharge of the recipient.[32](#)

Some 40 years after Backster first employed his crude polygraph mechanism to register the effect of thoughts, Korotkov verified those early discoveries with state-of-the-art equipment. He hooked up a potted plant to his GDV machine and asked his researchers to think of different emotions – anger, sadness, joy – and then positive and negative intentions towards the plant. Whenever a participant mentally threatened the plant, its energy field diminished. The opposite occurred if people approached the plant with water or feelings of love.

Largely because he lacked scientific credentials, Backster was never recognized for his contributions. He had stumbled across the first evidence that living things engage in a constant two-way flow of information with their environment, enabling them to register even the nuances of human thought. The more advanced scientific knowledge of physicists Fritz Popp and Konstantin Korotkov was needed to uncover the actual mechanism of that communication. Their research into the nature of quantum light emissions from living organisms suddenly made sense of Backster's findings. If thoughts are another stream of photons, it is perfectly plausible that a plant could pick up the signals and be affected by them.

The work of Backster, Popp and Korotkov suggested something profound about the effect of intention. Every last thought appeared to augment or diminish something else's light.

Notes – Chapter 3: The Two-Way Street

1. *For all history of Cleve Backster's discoveries and experiments, interview with Backster, October 2004 and his Primary Perception: Biocommunication with Plants, Living Foods, and Human Cells, Anza, Calif.: White Rose Millennium Press, 2003.*
2. *As Obi-Wan Kenobe tells Luke Skywalker, after Alderaan has been blown up by the Empire in Star Wars part IV: A New Hope: 'I feel a great disturbance in the Force. As if millions of voices suddenly cried out in terror, and were suddenly silenced.'*
3. *Presentation given at the Tenth Annual Parapsychology Association meeting in New York City, September 7, 1967. Also published as C Backster, 'Evidence of a primary perception in plant life', International Journal of Parapsychology, 1968; 10 (4): 329–48.*
4. *P. Dubrov and V. N. Pushkin, Parapsychology and Contemporary Science, New York and London: Consultants Bureau, 1982.*
5. *P. Tompkins and C. Bird, The Secret Life of Plants, New York: Harper & Row, 1973.*
6. *'Boysenberry to Prune, Boysenberry to Prune: Do you read me? Li detector expert Cleve Backster reported in the annual meeting of the American Association for the Advancement of Science that he had detected electrical impulses between two containers of yogurt at opposite ends of his laboratory. Backster claims the bacteria in the containers were communicating.' Esquire, January 1976.*
7. *Backster, 'Evidence of a primary perception', op. cit.*
8. *Backster, Primary Perceptions, op. cit.: 112–13.*

9. Backster, *Primary Perceptions*. See also Rupert Sheldrake, *Dogs That Know When Their Owners Are Coming Home and Other Unexplained Powers of Animals*, London: Three Rivers Press, 2000.
10. This and other personal details of events resulted from interviews with Ingo Swann, New York, July 2005.
11. See McTaggart, *The Field*, op. cit.: 39 for a full description of F.-A. Popp's earlier work.
12. All details of these experiments resulted from an interview between the author and Fritz-Albert Popp, January 2006.
13. R. M. Galle et al., 'Biophoton emission from *Daphnia magna*: A possible factor in the self-regulation of swarming', *Experientia*, 1991; 47: 457–60; R. M. Galle, 'Untersuchungen zum dichte und zeitabhängigen Verhalten der ultraschwachen Photonenemission von pathogenetischen Weibchen des Wasserfloh *Daphnia magna*.' Dissertation. Universität Saarbrücken, Fachbereich Zoologie, 1993.
14. F.-A. Popp et al., 'Nonsubstantial biocommunication in terms of Dicke's Theory', in M. W. Ho, F.-A. Popp and U. Warnke (eds.), *Bioelectrodynamics and Biocommunication*, Singapore: World Scientific Publishing, 1994: 293–317; J. J. Chang et al., 'Research on cell communication of *P. elegans* by means of photon emission', *Chinese Science Bulletin*, 1995; 40: 76–9.
15. J. J. Chang et al., 'Communication between Dinoflagellates by means of photon emission', in L. V. Belousov and F.-A. Popp (eds.), *Proceedings of International Conference on Non-equilibrium and Coherent Systems in Biophysics, Biology and Biotechnology*, Sep. 28–Oct. 2 1994, Moscow: Bioinform Services Co., 1995: 318–30.
16. Interview with Popp, Neuss, Germany, March 1, 2006.
17. F.-A. Popp et al., 'Mechanism of interaction between electromagnetic fields and living organisms', *Science in China (Series C)*, 2000; 43 (5): 507–18.
18. *Ibid.*
19. L. Belousov and N. N. Louchinskaia, 'Biophoton emission from developing eggs and embryos: Nonlinearity, wholistic properties and indications of energy transfer', in J. J. Chang et al. (eds.), *Biophotons*, London: Kluwer Academic Publishers, 1998: 121–40.
20. K. Creath and G. E. Schwartz, 'What biophoton images of plants can tell us about biofields and healing', *Journal of Scientific Exploration*, 2005; 19 (4): 531–50.
21. A. V. Tschulakow et al., 'A new approach to the memory of water', *Homeopathy*, 2005; 94: 241–7.
22. E. P. A. Van Wijk and R. Van Wijk, 'The development of a bio-sensor for the state of consciousness in a human intentional healing ritual', *Journal of International Society of Life Information Science (ISLIS)*, 2002; 20 (2): 694–702.

23. M. Connor, 'Baseline testing of energy practitioners: Biophoton imaging results.' Paper presented at the North American Research in Integrative Medicine conference, Edmonton, Canada, May 2006.
24. Personal details about K. Korotkov the result of multiple interviews with the author, November–March 2005–2006.
25. S. D. Kirlian and V. K. Kirlian, 'Photography and visual observation by means of high frequency currents', *J. Sci. Appl. Photogr.*, 1964; 6: 397– 403.
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CHAPTER FOUR

Hearts that Beat as One

NONE OF THE SCIENTISTS INVOLVED IN 'The Love Study' remember who came up with its name. It might have started as Elisabeth Targ's private joke, for the study involved couples who were installed in two different rooms and separated by a hallway, three doors, eight walls and several inches of stainless steel.[1](#)

The name was actually meant to be a gracious nod to the study's arcane benefactor, the Institute for Research on Unlimited Love at Case Western Reserve. As it happened, the study became a posthumous valentine to Targ, who was diagnosed with a fatal brain tumour just before the grant money came through. The Love Study would be a fitting tribute to Targ, as the first major scientific demonstration of exactly how intention physically affects its recipient, and the name proved especially apt in describing this process. When you send an intention, every major physiological system in your body is mirrored in the body of the receiver. Intention is the perfect manifestation of love. Two bodies become one.

Targ began her career as a mainstream psychiatrist, but made her name in 1999 with two remarkable studies at California Pacific Medical Center (CPMC) in San Francisco, which tested the possibility of remote healing with end-stage AIDS patients. Targ spent months designing her trial. She and her partner, psychologist and retired hospital administrator Fred Sicher, sought out a homogeneous group of advanced AIDS patients with the same degree of illness, including the same T-cell counts and number of AIDS-defining illnesses. Because they wished to test the effect of distant healing, and not any particular healing modality, they decided to recruit highly experienced, successful healers from diverse backgrounds who might represent an array of approaches.

Targ and Sicher gathered together an eclectic mix of healers from all across America – from orthodox Christians to Native American shamans – and asked them to send healing thoughts to a group of AIDS patients under strict double-blind conditions. All healing was to be done remotely so that nothing, such as the presence of a healer or healing touch, could confound the results. Targ created a strict double-blind rota: each healer received sealed packets with information about the patients to be healed, including their name, photo and T-cell counts. Every other week, the healers were assigned a new patient and asked to hold an intention for the health and well-being of the patient an hour a day for six days, with alternate weeks off for rest. In this manner, eventually every patient in the healing group would be sent healing by every healer in turn.

At the end of the first study, although 40 per cent of the control population died, all 10 of the patients in the treatment group were not only alive but far healthier in every regard.

Targ and Sicher repeated the study, but this time, doubled the size of their study population and tightened their protocol even further. They also widened their brief of the outcomes they planned to measure. In the second study, those sent healing were again far healthier on every parameter tested: significantly fewer AIDS-defining illnesses, improved T-cell levels, fewer hospitalizations, fewer visits to the doctor, fewer new illnesses, less severity of disease and better psychological well-being. The differences were decisive; for instance, the treatment group had six times fewer AIDS-defining illnesses and four times fewer hospitalizations at the end of the study than the controls.[2](#)

In Targ's original studies, the healing had been carried out by highly experienced, successful healers who had been chosen because they possessed a special gift. After the studies were completed, Targ grew interested in whether an ordinary individual could be similarly trained to use intention effectively.

For the Love Study, Targ found a sympathetic partner in Marilyn Schlitz, the vice president for research and education at the Institute of Noetic Sciences (IONS) The energetic blonde had a colourful national reputation because of her meticulously designed parapsychology studies and their spectacular results, which attracted the attention of the senior powers in consciousness research as well as the *New York Times*. During a long partnership with psychologist William Braud, Schlitz had conducted rigorous research into what became known in the psychic community as 'DMILS' – direct mental interaction with living systems – the ability of human thought to influence the living world around it.[3](#)

Throughout her career in parapsychology, Schlitz had been fascinated by remote influence; she was one of the first to examine the effect of intention in healing, and went on to assemble a vast database of healing research for IONS.

For the Love Study, Schlitz recruited Dean Radin, her IONS senior research and one of America's most renowned parapsychologists. Radin was to design both the study and some of its equipment; with his background in engineering and psychology he would ensure that both the study protocol and its technical detail were pristine. Targ enlisted Jerome Stone, a nurse and practising Buddhist who had worked with her on the AIDS studies, to design the intention programme and train the patients.

In 2002, after Targ died, Schlitz and the others vowed to carry on with the study and recruited Ellen Levine, one of Targ's colleagues from CPMC, to take her place and work with Stone as joint principal investigators.

The Love Study was to follow the basic study design of a perennial favorite among consciousness researchers: the sense of being stared at.[4](#)

In those studies, two people are isolated from each other in separate rooms and a video camera is trained on the receiver, who is also hooked up to skin conductance equipment, not unlike a polygraph machine – the type used in lie detection studies to detect an increase in ‘fight-or-flight’, unconscious autonomic nervous system activity. At random intervals, the ‘sender’ is instructed to stare at the subject on the monitor, while the ‘receiver’ is told to relax and try to think of anything other than the prospect of being stared at. A later comparison analysis determines whether the receiver’s autonomic system registered a reaction during those moments he or she was being stared at to determine whether the mere attention of the sender was unconsciously picked up by the most automatic systems of the receiver’s body.

Schlitz and Braud’s body of evidence on remote staring, conducted over 10 years, showed exactly such an effect. All the studies had been combined into a review that was published in a major psychology journal. The review concluded that the effects had been small but significant.[5](#)

The Love Study’s design was also inspired by the major DMILS studies conducted since 1963, which demonstrated that, under many types of circumstances, the electrical signalling in the brains of people gets synchronized.[6](#)

The frequencies, amplitudes and phases of the brain waves start operating in tandem. Although the studies followed slightly different designs, all of them asked the same question: can the stimulation of one person be felt in the higher central nervous system of another? Or, as Radin liked to think of it, after a sender gets pinched, does the receiver also feel the ‘ouch’?[7](#)

Two people wired up with a variety of physiological monitoring equipment, such as EEG machines, were isolated from each other in different rooms. One would be stimulated with something – a picture, a light or a mild electric shock. The researchers would then examine the two EEGs to determine if the receiver’s brain waves mirrored those of the sender when he or she was being stimulated.

The earliest DMILS research had been designed by psychologist and consciousness researcher Charles Tart, who carried out a series of brutal studies to determine whether people could empathetically feel another person’s pain. He administered shocks to himself, while a volunteer, isolated in a different room and hooked up to an array of medical gadgetry, was being monitored to see if his sympathetic nervous system somehow picked up Tart’s reactions. Whenever Tart jolted himself, the receiver registered an unconscious empathetic response in decreased blood volume and increased heart rate – as though he were also getting the shocks.[8](#)

Another fascinating early study had been carried out with identical twins. As soon as one twin closed his eyes and his brain electrical rhythms slowed to alpha waves, the

other twin's brain also slowed, even though his eyes were wide open.[9](#)

Harald Walach, a German scientist at the University of Freiburg, tried an approach that was guaranteed to magnify the sender's effects, in order to maximize the response in the receiver. The sender was shown an alternating black-and-white checkerboard, called a 'pattern reversal', which is known to trigger predictable, high-amplitude electrical brain waves in viewers. At the same instant, the EEG of the distant, shielded receiver recorded identical brain-wave patterns.[10](#)

Neurophysiologist Jacobo Grinberg-Zylberbaum, of the National Autonomous University of Mexico in Mexico City, had used this same protocol a decade before Walach but with a different twist: with light flashes rather than patterns as the stimulus. In this study, the particular patterns of firing in the brain of the sender, evoked by the light, turned out to be mirrored in the brain of the receiver, who was sitting in an electrically shielded room 14.5 metres away. Grinberg-Zylberbaum also discovered that an important condition determined success: the synchrony only occurred among pairs of participants who had met and established a connection by spending 20 minutes with each other in meditative silence.[11](#)

In earlier work, Grinberg-Zylberbaum had discovered that brainwave synchrony occurred not only between two people, but between both hemispheres of the brains of both participants, with one important distinction: the participant with the most cohesive quantum wave patterns sometimes set the tempo and tended to influence the other. The most ordered brain pattern often prevailed.[12](#)

In the most recent DMILS study, in 2005, a group of researchers from Bastyr University and the University of Washington gathered 30 couples with strong emotional and psychological connections and also a great deal of experience in meditation. The pairs were split up and placed in rooms 10 meters away from each other, with an EEG amplifier wired up to the occipital (visual) lobe of the brain of each participant. The moment each sender was exposed to a flickering light, he attempted to transmit an image or thought about the light to the partner. Of the 60 receivers tested, 5 of them, or 8 per cent, were shown to have significantly higher brain activation during times their partner 'sent' their visual images.[13](#)

The Washington researchers then selected five pairs of the participants who had scored a significant result, wired them up to a functional MRI, which measures minuscule changes in the brain during critical functions, and asked them to repeat the experiment. During the times the thought was 'transmitted', the recipients experienced an increase in blood oxygenation in a portion of the visual cortex of the brain. This increase did not occur when the sending partner was not being visually stimulated.[14](#)

The Bastyr researchers replicated their study, this time with volunteers highly experienced in meditation, and got some of the strongest correlations between senders and receivers of all the studies thus far.

The Bastyr study represented a major breakthrough in research on direct mental influence. It demonstrated that the brain-wave response of the sender to the stimulus is mirrored in the receiver, and that the stimulus in the receiver occurs in an identical place in the brain as that of the sender. *The receiver's brain reacts as though he or she is seeing the same image at the same time.*

A final extraordinary study examined the effect of powerful emotional involvement on remote influence. Researchers at the University of Edinburgh studied and compared the EEGs of bonded couples, matched pairs of strangers, and several individuals with no partner but who nevertheless thought they were being paired off and having their brain waves compared. Everyone who had been paired off, whether he knew his partner or not, displayed increased numbers of brain waves in synchrony. The only participants who did not demonstrate this effect were those who had no partner.[15](#)

Radin carried out a variation of this experiment, attaching pairs who had close bonds – couples, friends, parents and their children. In a significant number of instances, the EEGs of the senders and receivers appeared to synchronize.[16](#)

In designing the Love Study, Schlitz and Radin also had been influenced by other research showing that, during acts of remote influence, the recipient's EEG waves mirror those of the sender. In a number of studies of healing, the EEG waves of the patient synchronize with those of the healer during moments when healing energy is being 'sent'.[17](#)

Brain mapping during certain types of healing, such as bioenergy, also shows evidence of brain-wave synchrony. [18](#)

In many instances, when one person is sending focused intention to another, their brains appear to become entrained.

Entrainment is a term in physics which means that two oscillating systems fall into synchrony. It was coined in 1665 by the Dutch mathematician Christiaan Huygens, after discovering that two of his clocks with pendulums standing in close approximation to each other had begun to swing in unison. He had been toying with the two pendulums and found that even if he started one pendulum swinging at one end, and the other at the opposite end, eventually the two would swing in unison.

Two waves peaking and troughing at the same time, are considered 'in phase', or operating in synch. Those peaking at opposite times are 'out of phase'. Physicists

believe that entrainment results from tiny exchanges of energy between two systems that are out of phase, causing one to slow down and the other to accelerate until the two are in phase. It is also related to resonance, or the ability of any system to absorb more energy than normal at a particular frequency (the number of peaks and troughs in one second). Any vibrating thing, including an electromagnetic wave, has its own preferential frequencies, called 'resonant frequencies', where it finds vibrating the easiest. When it 'listens' or receives a vibration from somewhere else, it tunes out all pretenders and only tunes into its own resonant frequency. It is a bit like a mother instantly recognizing her child from among a mass of school children. Planets have orbital resonances. Our sense of hearing operates through a form of entrainment: different parts of a membrane of the inner ear resonate to different frequencies of sound. Resonance even occurs in the seas, such as in the tidal resonance of the Bay of Fundy in the northeast end of the Gulf of Maine, near Nova Scotia.

Once they march to the same rhythm, things that are entrained send out a stronger signal than they do individually. This most commonly occurs with musical instruments, which sound amplified when all playing in phase. At the Bay of Fundy, the time required for a single wave to travel from the bay's mouth to its opposite end and back is exactly matched by the time of each tide. Each wave is amplified by the rhythm of each tide, resulting in some of the highest tides in the world.

Entrainment also occurs when someone sends a strong intention to cause harm, which became evident in the *tohate* experiments of Mikio Yamamoto of the National Institute of Radiological Sciences in Chiba and the Nippon Medical School in Tokyo. *Tohate* is a kind of mental stand-off between two *Qigong* practitioners, one of whom receives a sensory shock and is eventually made to submit and move back several yards without any physical contact from the other. The central question posed by the technique, in Yamamoto's mind, was whether the effect of *tohate* is psychological or physical: does the opponent move back because of psychological intimidation, or is he knocked over by the *qi* of his opponent?

In the first of Yamamoto's studies, a *Qigong* master was isolated in an electromagnetically shielded room on the fourth floor of a building, while his student was similarly isolated on the first floor.

Yamamoto signalled for the master to perform '*qi* emission' over 80 seconds at random intervals. Each time, he tracked their separate movements – the sending of the *qi* and the start of the pupil's recoil. In nearly a third of the 49 such trials – a highly significant result – whenever the master engaged in *tohate* movements, his opponent in the other room was physically knocked back. In a second set of 57 trials, Yamamoto wired both teacher and pupil to EEG machines. Whenever the master emitted *qi*, his pupil showed an increase in the number of alpha brain waves in his

right frontal lobe, suggesting that this was where the body initially receives the intention 'message'.

Yamamoto's final set of trials examined the EEG-recorded brain waves of both master and student.

Whenever the master performed *tohate*, the beta brain waves of both men demonstrated a greater sense of coherence.[19](#)

In an earlier study carried out by the Tokyo group, the brain waves of the receiver and sender became synchronized within one second during *tohate*.[20](#)

Besides resonance, the DMILS studies offered evidence of another phenomenon during intention: the receiver anticipated the information by registering the 'ouch' a few moments *before* the pinch occurred in the sender.

In 1997, in his former laboratory at the University of Nevada, Radin discovered that humans may receive a physical foreboding of an event.

He set up a computer that would randomly select photos designed to calm, to arouse, or to upset a participant. His volunteers were wired to physiological monitors that recorded changes in skin conduction, heart rate and blood pressure, and they sat in front of a computer that would randomly display colour photos of tranquil scenes (landscapes), or scenes designed to shock (autopsies) or to arouse (erotic materials).

Radin discovered that his subjects were registering physiological responses *before* they saw the photo. As if trying to brace themselves, their responses were highest before they saw an image that was erotic or disturbing.

This offered the first laboratory proof that our bodies unconsciously anticipate and act out our own future emotional states and that the nervous system does not merely cushion itself against a future blow, but also works out the emotional meaning of it.[21](#)

Dr Rollin McCraty, executive vice-president and director of research for the Institute of HeartMath, in Boulder Creek, California was fascinated by the idea of shared physical foreboding of an event, but wondered where exactly in the body this intuitive information might first be felt. He used the original design of Radin's study with a computerized system of randomly generated arousing photos, but hooked up his participants to a greater complement of medical equipment.

McCraty discovered that these forebodings of good and bad news were felt in both the heart and brain, whose electromagnetic waves would speed up or slow down just before a disturbing or tranquil picture was shown. Furthermore, all four lobes of the cerebral cortex appeared to take part in this intuitive awareness. Most astonishing of

all, the heart appeared to receive this information moments *before* the brain did. This suggested that the body has certain perceptual apparatus that enables it continually to scan and intuit the future, but that the heart may hold the largest antenna. After the heart receives the information, it communicates this information to the brain.

McCraty's study had shown certain fascinating differences between the sexes. Both the heart and brain became entrained with each other earlier and more frequently in women than they did in men. McCraty concluded that this offered scientific evidence of the universal assumption that women are naturally more intuitive than men and more in touch with their heart centre.[22](#)

McCraty's conclusion – that the heart is the largest 'brain' of the body – has now gained credibility after research findings by Dr John Andrew Armour at the University of Montreal and the Hôpital du Sacré-Coeur in Montreal.

Armou discovered neurotransmitters in the heart that signal and influence aspects of higher thought in the brain.[23](#) McCraty discovered that touch and even mentally focusing on the heart cause brain-wave entrainment between people.

When two people touched while focusing loving thoughts on their hearts, the more 'coherent' heart rhythms of the two began to entrain the brain of the other.[24](#)



When two people touched while focusing loving thoughts on their hearts, the more 'coherent' heart rhythms of the two began to entrain the brain of the other.

Armed with this new evidence about the heart, Dean Radin and Marilyn Schlit decided to explore whether remote mental influence extended to anywhere else in the body. An obvious place to explore was the gut. People speak about intuition as a

'gut instinct' or 'gut feeling'. Certain researchers have even referred to the gut as a 'second brain'.[25](#)

Radin wondered if a gut instinct was accompanied by an actual physical effect.

Radin and Schlitz gathered 26 student volunteers, paired them, and this time wired them up to an electrogastrogram (EGG), which measures the electrical behaviour of the gut; monitors on the skin usually closely match the frequencies and contractions of the stomach. Although the Freiburg study had shown otherwise, Radin and Schlitz believed that familiarity could only help to magnify the effects of remote influence. In case some sort of physical connection was indeed important, Radin asked all the participants to exchange some meaningful object first.

Radin put one participant from a pair in one room. The other sat in another, darkened room, attached to an electrogastrogram, viewing live video images of the first person. Images periodically flashed on another monitor, accompanied by music designed to arouse particular emotions: positive, negative, angry, calming or just neutral.

The results revealed another example of entrainment – this time in the gut.

The EGG readings of the receiver were significantly higher and correlated with those of the sender when the sender experienced strong emotions, positive or negative. Here was yet more evidence that the emotional state of others is registered in the body of the receiver – in this case, deep in the intestines – and that the home of the gut instinct is indeed the gut itself.[26](#)

This latest evidence was further proof that our emotional responses are constantly being picked up and echoed in those closest to us.[27](#) In every one of these studies, the bodies of the pairs had become entrained or 'entangled' as Radin called it;[28](#) the recipients were 'seeing' or feeling what their partners actually saw or felt, in real time.

As this research intimates, intention might be an attunement of energy. The DMILS research established that, under certain conditions, the heart rate, the arousal of the autonomic nervous system, the brain waves and the blood flow to the extremities of different people all become entrained, even when they are situated at a distance. Nevertheless, in most of the DMILS studies, the correlated responses resulted from a simple stimulation of the sender, which the recipient unconsciously picked up. Except for one instance, no one attempted to influence another person.

Schlitz and Radin now wanted to find out whether they would achieve similar correlations if the sender were actually sending an intention to heal. For the Love Study, Schlitz and her colleagues decided to recruit ordinary individuals and train

them in healing techniques. They wondered whether certain conditions were more favourable than others for achieving entrainment.

Many healing studies intimated that motivation, interpersonal connection and a shared belief system were vital to success.

Grinberg-Zylberbaum believed that a 'transferred potential', as he termed this form of entrainment, occurred only among those who had undergone some meditative regime and then only after some sort of psychic connection between sender and receiver had been established. Nevertheless, in the Freiberg study, many of the pairs had never met each other and had not had a chance to establish a bond.

The German researchers had concluded that 'connectedness' and mental preparation may play a role, but were not crucial. In Schlitz's view, motivation was a key component of success. The more urgent the situation, such as would occur with a partner suffering from cancer, the more motivated his or her partner would be in attempting to get him or her well.

Schlitz and her fellow researchers decided to seek out couples with a wife suffering from breast cancer, and began advertising around the San Francisco Bay Area for volunteers.

It soon became apparent that they would have to widen their original brief. The breast-cancer population of the Bay Area, which is higher than average in the USA, has been extremely well studied.

From the lack-luster response to their advertising, it appeared that sufferers were unwilling to take part in yet more research. The scientists decided to open the study to any couple if either partner were suffering from cancer of any variety.

Eventually 31 couples volunteered, including healthy couples who were to act as controls.

Jerome Stone wrote a training manual for the couples, after analysing a number of healers and distilling their common practices.[29](#)

The first component of his programme involved teaching the sender how to focus and concentrate, as occurs in meditation, to create a high degree of sustained *attention*. The scientific evidence demonstrates that meditation establishes more coherent brain waves; at least 25 studies show that EEG synchronization occurs between the four regions of the brain during meditation.[30](#)

Other studies of meditation have shown that it creates more coherent biophoton emissions[31](#) and in general aids healing.

Stone also believed that his senders needed to learn how to generate *compassion* or empathy for their partners, with a technique based largely on the Tonglen Buddhist idea of 'giving and receiving'. This practice would train the partner to develop a true understanding of the suffering of another, to take on the suffering without being burdened by it, and to transform it through the process of sending healing.

Developing true empathy would also help to dissolve the boundaries and sense of self between the sender and receiver. Positive, loving thoughts also had positive physiological effects. Rollin McCraty's research at HeartMath showed that a steady (or, as they called it, 'coherent') variation in heartbeat was more likely with 'positive' – loving or altruistic – thoughts and that this 'coherence' was quickly picked up by the brain, which soon pulsed in synchrony³² and evidenced improved cognitive performance.³³

After Stone instructed the partners in simple techniques of meditation, he also taught them to be compassionate when carrying out intention. The final aspect of Stone's training involved instilling *belief* and confidence in both senders and receivers.

Stone had discovered evidence in both the healing and parapsychological literature that belief in the process assists in the success of psychic processes such as ESP, which, like intention, involves 'transferring' information across distance.³⁴

Although the training programme was originally intended to run for eight weeks, limited funding meant that Stone had to compress his workshop into a single day, to be followed up with homework and practice.

Radin divided the couples into three groups.

The **first group** (the 'trained group') was to undergo Stone's training, practise compassionate intention daily for three months and then carry out the test.

The **second group** (called the 'wait group') was to carry out the test first and then have the training.

The 18 healthy couples comprising the **third group** (the control group) was to have no training at all, but simply undergo the test.

With all three groups, the member of the couple with the cancer (or one of the designated partners in the control group) was asked to sit in a black reclining chair placed in a one-ton, solid steel, double-walled, electromagnetically shielded enclosure.

The tiny Lindgren/ETS chamber was separated from the outside world by two layers of steel and one of solid wood, which blocked out all sound and all electromagnetic

energy. Any electrical signals were carried out of the chamber by a fibre-optic cable, to ensure that the room remained, electromagnetically speaking, a solitary confinement.

Each inhabitant was fitted to an array of medical gadgetry to measure brain waves, heartbeat, breathing rate, skin conductance and peripheral blood flow. A video camera stood discretely in the corner.

The room was curtained in earth tones and furnished with soft table lighting and an artificial, floor-to-ceiling weeping fig tree. When the room was occupied, ambient music flooded the space. The furnishings and music, and even a large colour poster of a cascading mountain stream, were all intended to distract from the fact that once the 400-pound steel door with an articulated closing mechanism snapped shut, the inhabitant was essentially trapped inside the warmer equivalent of a meatpacking-plant refrigerator.

Some 20 metres away, the other partner was seated in the dark, attached to the same medical equipment as his or her partner, staring at a small blank TV screen. Bunched towels blocked out the last vestiges of light. Whenever the image of the partner in the refrigerator room abruptly flashed on the television screen, the other member of the couple was to send a compassionate intention to his or her partner for 10 seconds.

Stone, Radin and their colleagues planned to examine two different outcomes: whether the training improved the marriage, and also whether there was any correspondence between the physical sensations of sender and receiver. Although they hoped to examine whether the intentions sent also affected the medical prognosis, limited funding made that aspect of the study impossible.

Stone and Levine were given the task of analysing the social aspects of the study. Initially they discovered that the training made no difference to the quality of the couples' marriages.

The finding was not altogether surprising, considering that anyone prepared to be part of a study involving three months of training was already likely to be extremely committed to the partnership. And Schlitz had aimed to recruit motivated partners when she designed the study. A later, more detailed analysis of the figures showed that the intention training and practice had indeed improved the couples' marriages, but Radin concluded that these effects were due to their expectation of improved relations.

Then Radin compiled all the physiological data from the three groups and studied the results between partners and group composite averages. Each physiological

response offered fascinating information about the effect of intention on the receiver. For instance, in the case of measurements of blood to the extremities, in every group, the sender's skin conductance increased 2 seconds after seeing the partner's image, and the receiver recorded a similar arousal a half second after the image had flashed.

However, unlike the earlier DMILS studies, where the skin conduction response in the receiver resembled that of a 'startle reflex' and quickly tailed off, in this instance the response persisted 7 seconds after the stimulus.

The receiver clearly appeared to be responding to intention – indeed, almost instantaneously.

In fact, the receiver's response occurred at least 1 second faster than it would have been possible for the sender to have consciously formulated an intention. Radin was not sure whether this meant that the receiver had had a premonition of the intention.

It might simply have reflected the turgid nature of the skin conductance response; the receiver was likely responding in his or her extremities to information sent by the sender's central nervous system, which would have reacted to the initial stimulation of the image on the monitor far more quickly than the electrical impulses sent to his or her fingertips. Nevertheless, in Radin's view, the two skin conductance responses were tracking each other, even if they were slightly out of phase.

A similar situation occurred with the heart rate. The sender's heart rate increased 5 seconds after the stimulus prompt to send the intention – which was consistent with the physical response that occurs in the body during the process of making some sort of mental effort. But an identical increase took place in the receiver, which would not happen ordinarily if he or she were simply resting in a recliner.

Blood flow followed a similar pattern. Whenever we experience something that stimulates us, the vascular network in our extremities constricts slightly, to maximize blood flow to the core of the body. In the Love Study, this phenomenon occurred in the sender, and was soon imitated in the body of the receiver.

As for respiration, on average, whenever the stimulus image appeared, the sender immediately inhaled sharply and blew out the air 15 seconds later. This respiratory response resembles that of someone about to steady himself for the task at hand. In this case, Radin witnessed a different response in the receiver. During the first 5 seconds, the receiver's respiration faltered, almost as though he or she had stopped breathing, and then resumed with a large exhale in the final 5 seconds of the intention. It was as though the receiver had been listening with care, holding her breath and straining to hear something, before sighing with relief as soon as the stimulation had passed.

But it was the brain-wave results that proved to be the most interesting. Whenever the receiver's image flashed on the screen, the senders recorded a little upturn in brain waves, like a 'flinch response', and then a huge spike for about a third of a second before they dropped sharply and took about one second to come back to baseline. In the sender, this tiny initial upturn represents something called a P300 wave – a well-established phenomenon that records the time that the brain takes to process the switching on of a light. The drop represents the time it takes for internal attention to modulate the stimulus into a response.

In this instance, the receivers had no P300 wave, but their brain waves nevertheless mimicked the virtually vertical plunge of the brain wave that shortly followed in the sender, even though, unlike the sender, the receiver had had no stimulus. The brain of the receiver was reacting just as it does when asleep and dreaming. The receivers had registered an emotional reaction, even though there was no tangible stimulus.

Radin's results were all the more remarkable because the receivers had not been told how long the stimulus period would be, and neither senders nor receivers knew in advance how long the sender would have to wait before the partner's image flashed on screen. A computer program randomly selected the time frame, which ranged from 5 to 40 seconds. This meant that any expectation on the part of either member of the couples could not explain the results.

Radin then compared the responses of the groups. All three groups had shown an effect. In every instance, each physiological response of the receivers had tracked those of the senders. However, the most prolonged pattern occurred among the cancer patients whose partners had been trained in compassionate intention.

The receivers in the training group not only responded to the stimulus, but also kept responding over 8 of the 10 seconds of the intention. In quantum terms, the couples had become as one.[35](#)

The Love Study indicates a number of profound suggestions about the nature of intention.

Sending a directed thought seems to generate a palpable energy; whenever one of Radin's senders sent a healing intention, many subtle aspects of the receiver's body became activated, as though he had received a minuscule electric shock. It seemed to be a kind of activating awareness, as though his body had *felt or heard* the healing signal.

There had even been an element of anticipation in the receiver; some of the physiological reactions recorded suggested that the receiver had felt the partner's healing intention *before* he had even sent it.

People appear to receive healing deep in their bodies by being retuned to the more coherent energy of the healer's intention. During healing, it could be that the 'orderly' energy of the well person entrains and 're-orders' the sick.

In order to have the most powerful effect, a healer or sender needs to become 'ordered' on some subatomic level, mentally and emotionally. The Love Study demonstrates that certain conditions and mental states make our intention especially powerful and ourselves more ordered, and that these states can be achieved with training. The success of the basic training programme that Schlitz, Radin and Stone assembled suggests that attention, belief, motivation and compassion are important for intention to work, but there are probably other conditions that intensify its effects.

I needed, for instance, to find out how we can loosen our psychological boundaries. It was becoming clear to me: when we send intention, in a manner of speaking, we have to 'become' the other.

Notes – Chapter 4: Hearts that Beat as One

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PART TWO

Powering Up

For every atom belonging to me as good belongs to you.
- 'Song of Myself', Walt Whitman

CHAPTER FIVE

Entering Hyperspace

IN A DRAUGHTY MONASTERY high in the Himalayas in northern India during the winter of 1985, a group of Tibetan Buddhist monks were seated quietly, deep in meditation. Although scantily clad, they appeared oblivious to the chilly indoor air temperature, which approached freezing. A fellow monk passed between them, draping each, in turn, with sheets drenched with cold water. Such extreme conditions would ordinarily shock the body and send the core temperature plummeting. If body temperature falls by only 7°C, within minutes a person will lose consciousness and all vital signs.

Instead of shivering, the monks began to sweat. Steam rose from the wet sheets; within an hour, they were thoroughly dry. The attendant replaced the dry sheets with new ones, also drenched in ice-cold water. By this time, the monks' bodies had become the equivalent of a furnace. Those sheets were efficiently dried, as was a third batch.

A team of scientists led by Herbert Benson, a cardiologist at Harvard Medical School, stood nearby, examining an array of medical equipment to which they had attached the monks for any clues as to what particular physiological mechanism might have enabled the body to generate this extraordinary level of heat.

For a number of years, Benson had explored the effects of meditation on the brain and the rest of the body. He'd embarked on an ambitious research programme, studying Buddhists in various remote outposts around the world who had spent many

years in disciplined practice. During one trip to the Himalayas, he also videotaped monks, dressed only in light shawls, as they spent a freezing February night outdoors on a mountain ledge 4600 meters above sea level. Benson's film showed that they had slept soundly through the night, without clothing or shelter.

In his travels, Benson had witnessed many extraordinary feats of intention – mastery over temperature or metabolic rate that could even produce a state resembling hibernation. The monks monitored by Benson's team had raised the temperature of their extremities by up to 9.4°C and lowered their metabolism by more than 60 per cent.[1](#)

Benson realized that this represented the largest variation in resting metabolism ever reported. During sleep, by contrast, metabolism only drops by 10 to 15 per cent; even experienced meditators can only decrease it by 17 per cent, at best. But that day in the Himalayas, he had observed the impossible in terms of mental influence. The monks had used their bodies to boil freezing water simply through the power of their thoughts.[2](#)

Benson's enduring enthusiasm for meditation ignited interest at major academic institutions across America. By the end of the twentieth century, monks had become the favourite guinea pigs of the neuroscience laboratory. Scientists from Princeton, Harvard, the University of Wisconsin and the University of California–Davis followed Benson's lead by wiring up monks to state-of-the-art monitoring equipment and studying the effects of intensive, advanced meditation. Entire conferences were held on meditation and the brain.[3](#)

It was not the practice itself that fascinated these scientists, but its effect on the human body, particularly the brain, and the possibilities this suggested. By studying the biological effects in such detail, scientists hoped to understand the neurological processes that occur during feats of highly directed thought, as the monks had displayed in the Himalayas.

Monks also offered scientists an opportunity to study whether years of focused attention stretch the brain beyond its usual limits. Did the brain of a monk become the equivalent of an Olympic athlete's body – more highly developed and ultimately transformed after gruelling discipline and practice?

Do training and experience change the physiology of the brain over time? Would practice enable you to become a bigger and better transmitter of intention? The answers would in turn address a long-standing debate in neuroscience: is neural structure basically hard-wired from youth or plastic – changeable – depending on the nature of a person's thoughts through life?

For me, the most intriguing question about this research on focused attention was the means by which a Buddhist monk could turn himself into a human boiler, and how these means compared with techniques and practices of other ancient traditions. Like Benson, I was intrigued by 'masters' of intention: practitioners of ancient disciplines Buddhism, *Qigong*, shamanism, traditional native healing – who had been trained to perform extraordinary acts through their thoughts. I wanted to work out the common denominators they shared.

Do the steps taken by a *Qigong* master to send *Qi* resemble those of a Buddhist monk during meditation?

Which mental disciplines ensure that a healer will enter a state enabling him to repair another person's body?

Are 'masters' of intention graced with special neurological gifts that enable them to use their minds more powerfully than the rest of us, or did they acquire a skill that ordinary people could learn as well? And, perhaps most important, what did the neurological study of monks tell me about the effect of focused intention on the brain? Would practice enable you to become a bigger and better transmitter of intention?

I began studying scientific research about healing methods from a variety of traditions and then conducted my own questionnaire and interviews with healers and 'master' intenders of all persuasions.[4](#)

I was aided in my research by the work of psychologist Stanley Krippner and his student Allan Cooperstein at Saybrook Graduate School. A clinical and forensic psychologist, Cooperstein had conducted a thorough study of the various techniques used by distant healers for his doctoral thesis, including an analysis of scholarly books on healing and exhaustive written and verbal interviews with well-known practitioners who had scientific evidence of success in healing.[5](#)

In every instance, I discovered, the most important first step involved achieving a state of concentrated focus, or *peak attention*.

According to Krippner, an expert on shamanic and other native traditions, virtually all native cultures carry out remote healing during an altered state of consciousness and achieve a state of concentrated focus through a variety of means.[6](#)

Although the use of hallucinogenic drugs such as ayahuasca is common, many cultures use a strong repetitive rhythm or beat to create that state; the Native American Ojibway *wanbeno*, for instance, use drumming, rattling, chanting, naked dancing and handling of live coals.[7](#)

Drumming is particularly effective in producing a highly concentrated focus; a number of studies have shown that listening to the beat of a drum causes the brain to slow down into a trancelike state.[8](#)

As Native Americans discovered, even intense heat, as in a sweat lodge, can transport individuals to an altered state.

In my own study of intention ‘masters’, I spoke with Bruce Frantzis, arguably the greatest *Qigong* master in the West. A martial arts champion, with black belts in five Japanese martial arts, he also learned healing *Qigong* through years of study with Chinese masters.

Frantzis’s powers of intention were legendary; he had been videoed sending people flying across the room simply by directing *Qi*. In his fighting days, he had put several people into wheelchairs. Now, knowing its extraordinary power, he reserved *Qi* for healing. During my own meeting with him, Frantzis gave a short demonstration of the power of directed *Qi*. After a moment of intense concentration, the plates of his skull began to undulate over the top of his head like a rolling surf.[9](#)

Frantzis taught his students how to develop peak attention gradually, through intense concentration on their breathing. Although they began with very short bursts of ‘longevity’ breathing, they would work on extending these periods until eventually they could hold this focus continuously. They would also be taught methods of becoming acutely aware of all physical sensation.[10](#)

The healers I interviewed entered this focused state through a variety of means: meditation; prayer; intense attention on the person to be healed; symbolic or mythic ideas; strong mental images of a situation producing the desired change; verbal affirmations; mental imagery; even internal autosuggestions as a warm-up exercise. One healer established focused attention by saturating his awareness with the goal that he was trying to achieve.

Dr Janet Piedilato, a shamanic healer, will often ‘gently hum or chant’ or use a ‘rattle or other instrument’. Dr Constance Johnson, a Reiki practitioner, can return to an altered state at will. Others need to work hard to achieve this transformation: The Reverend Francis Geddes, a spiritual healer, will meditate on a small object like a pebble, leaf, or twig in a ‘very concentrated manner for ten minutes’.

Still others use the patient as the object of meditation. As Dr Judith Swack, a mind–body healer who has developed her own holistic psychotherapy system, says: ‘I look directly at the client and focus all of my senses forward toward the client and enter a receptive state where I pay internal attention to any subtle information and impression coming in like a kind of radar.’ Many other healers likewise enter an altered state,

simply by 'listening to the patient' – 'audibly or otherwise'. 'Just thinking of the need to help someone,' wrote Dr Piedilato, 'slows the blood in my veins.'

Initially, many healers experience a heightening of their cognitive processes, but most soon reach a point when inner chatter ceases, and they experience a falling away of all sensation but pure image. The focusing seems to dissolve their own boundaries. They suddenly become aware of the inner workings of the patient's body and ultimately have a sense of being engulfed by the healee.

I was especially interested in the effect of this intense concentration on the activity of the brain. Does the brain slow down or speed up? The received wisdom is that during meditation the brain slows down. The bulk of the research examining the electrical activity of the brain during meditation indicates that meditation leads to a predominance of either alpha rhythms (slow, high-amplitude brain waves with frequencies of 8–13 hertz, or cycles per second), which also occurs during light dreaming, or even the slower theta waves (4–7 hertz), which typify the state of consciousness during deep sleep.¹¹ During ordinary waking consciousness, the brain operates much faster, using beta waves (around 13–40 hertz). For decades, the prevailing view has been that the optimum state for manifesting intention is an 'alpha' state.

Richard Davidson, a neuroscientist and psychologist at the University of Wisconsin's Laboratory for Affective Neuroscience, recently put this view to the test. Davidson was an expert in 'affective processing' – the place where the brain processes emotion and the resulting communication between the brain and body. His work had come to the attention of the Dalai Lama, who invited him to visit Dharamsala, India, in 1992; a science buff, his Holiness wished to understand more about the biological effects of intensive meditation.

Afterwards, eight of the Dalai Lama's most seasoned practitioners of Nyingmapa and Kagyupa meditation were flown to Davidson's lab in Wisconsin.

There, Davidson attached 256 EEG sensors to each monk's scalp in order to record electrical activity from a large number of different areas in the brain.

The monks were then asked to carry out compassionate meditation. As with Jerome Stone's intention regime, the meditation entailed focusing on an utter readiness to help others and a desire for all living things to be free of suffering.

For the control group, Davidson enlisted a group of undergraduates who had never practiced meditation and arranged for them to undergo a week's training, then attached them to the same number of EEG sensors to monitor their brains during meditation.

After 15 seconds, according to the EEG readings, the monks' brains did not slow down; they began speeding up.

In fact, they were activated on a scale neither Davidson nor any other scientist had ever seen. The monitors showed sustained bursts of high gamma-band activity – rapid cycles of 25–70 hertz. The monks had rapidly shifted from a high concentration of beta waves to a preponderance of alpha, back up to beta and finally up to gamma.

Gamma band, the highest rate of brain-wave frequencies, is employed by the brain when it is working its hardest: at a state of rapt attention, when sifting through working memory, during deep levels of learning, in the midst of great flashes of insight.

As Davidson discovered, when the brain operates at these extremely fast frequencies, the phases of brain waves (their times of peaking and troughing) all over the brain begin to operate in synchrony. This type of synchronization is considered crucial for achieving heightened awareness.[12](#)

The gamma state is even believed to cause changes in the brain's synapses – the junctions over which electrical impulses leap to send a message to a neuron, muscle or gland.[13](#)

That the monks could achieve this state so rapidly suggested that their neural processing had been permanently altered by years of intensive meditation.

Although the monks were middle-aged, their brain waves were far more coherent and organized than those of the robust young controls. Even during their resting state, the Buddhists showed evidence of a high ratio of gamma-band activity, compared with that of the neophyte meditators.

Davidson's study bolstered other pieces of preliminary research suggesting that certain advanced and highly focused forms of meditation produce a brain operating at peak intensity.[14](#)

Studies of yogis have shown that, during deep meditation, their brains produce bursts of high-frequency beta or gamma waves, which often are associated with moments of ecstasy or intense concentration.[15](#)

Those who can withdraw from external stimuli and completely focus their attention inward appear more likely to reach gamma-wave hyperspace. During peak attention of this nature, the heart rate also accelerates.[16](#)

Similar types of effects have been recorded during prayer. A study monitoring the brain waves of six Protestants during prayer found an increase in brain-wave speed

during moments of the most intense concentration.[17](#)

Different forms of meditation may produce strikingly different brain waves. For instance, yogis strive for *anuraga*, or a sense of constant fresh perception; Zen Buddhists aim to eliminate their response to the outer world. Studies comparing the two find that the former produces heightened perceptual awareness – magnified outer focus – while the latter produces heightened inner absorption – magnified inner awareness.[18](#)

Most research on meditation has concerned the type that focuses on one particular stimulus, such as the breath or a sound, like a mantra. In Davidson's study, the monks concentrated on having a sense of compassion for all living things. It may be that compassionate intention – and other similar, 'expansive' concepts – produces thoughts that send the brain soaring into a supercharged state of heightened perception.

When Davidson and his colleague Antoine Lutz wrote up their study, they realized that they were reporting the highest measures of gamma activity ever recorded among people who were not insane.[19](#)

In their results they noticed an association between level of experience and ability to sustain this extraordinarily high brain activity; those monks who had been performing meditation the longest recorded the highest levels of gamma activity. The heightened state also produced permanent emotional improvement, by activating the left anterior portion of the brain the portion most associated with joy.

The monks had conditioned their brains to tune into happiness most of the time.

In later research, Davidson demonstrated that meditation alters brain-wave patterns, even among new practitioners. Neophytes who had practised mindfulness meditation for only eight weeks showed increased activation of the 'happy-thoughts' part of the brain and enhanced immune function.[20](#)

In the past, neuroscientists imagined the brain as something akin to a complex computer, which got fully constructed in adolescence. Davidson's results supported more recent evidence that the 'hardwired' brain theory was outdated.

The brain appeared to revise itself throughout life, depending on the nature of its thoughts. Certain sustained thoughts produced measurable physical differences and changed its structure. Form followed function; consciousness helped to form the brain.

Besides speeding up, brain waves also synchronize during meditation and healing. In fieldwork with indigenous and spiritual healers in five continents, Krippner suspected

that, prior to healing, the healers all underwent brain 'discharge patterns' that produce a coherence and synchronization of the two hemispheres of the brain, and integrate the limbic (the lower emotional centre) with the cortical systems (the seat of higher reasoning).[21](#)

At least 25 studies of meditation have shown that, during meditation, EEG activity between the four regions of the brain synchronizes.[22](#)

Meditation makes the brain permanently more coherent – as might prayer. A study at the University of Pavia in Italy and the John Radcliffe Hospital in Oxford showed that saying the rosary had the same effect on the body as reciting a mantra. Both were able to create a 'striking, powerful, and synchronous increase' in cardiovascular rhythms when recited six times a minute.[23](#)

Another important effect of concentrated focus is the integration of both left and right hemispheres. Until recently, scientists believed that the two sides of the brain work more or less independently. The left side was depicted as the 'accountant', responsible for logical, analytical, linear thinking, and speech, and the right side, as the 'artist', providing spatial orientation, musical and artistic ability, and intuition.

But Peter Fenwick, consultant neuropsychiatrist at the Radcliffe Infirmary in Oxford St Thomas' Hospital, Bethlehem Hospital and the Institute of Psychiatry at the Maudsley Hospital, gathered evidence to show that speech and many other functions are produced in both sides of the brain and that the brain works best when it can operate as a totality. During meditation, both sides communicate in a particularly harmonious manner.[24](#)

Concentrated attention appears to enlarge certain mechanisms of perception, while tuning out 'noise'. Daniel Goleman, author of *Emotional Intelligence*,[25](#) carried out research showing that the cortices of meditators 'speed up', but get cut off from the limbic emotional center.

With practice, he concluded, anyone can carry out this 'switching-off' process, enabling the single mode of the brain to experience heightened perception without an overlay of emotion or meaning.[26](#)

During this process, all of the power of the brain is free to focus on a single thought: an awareness of what is happening at the present moment.

Meditation also appears to permanently enhance the brain's reception. In several studies, meditators have been exposed to repetitive stimuli like light flashes or clicks. Ordinarily, a person will get used to the clicks, and the brain, in a sense, will switch off

and stop reacting. But the brains of the meditators continued to react to the stimuli – an indication of heightened perception of every moment.[27](#)

In one study, practitioners of mindfulness meditation – the practice of bringing heightened, non-judgemental awareness of the senses' perceptions to the present moment – were tested for visual sensitivity before and immediately after a three-month retreat, during which time they had practiced mindfulness meditation for 16 hours a day.

The staff members who did not practice the meditation acted as a control group. The researchers were testing whether the participants could detect the duration of simple light flashes and the correct interval between successive ones.

To those without mental training in focusing, these flashes would appear as one unbroken light.

After the retreat, the practitioners were able to detect the single-light flashes and to differentiate between successive flashes.

Mindfulness meditation enables its practitioners to become aware of unconscious processes and to remain exquisitely sensitive to external stimuli.[28](#)

As these studies indicate, certain types of concentrated focus, like meditation, enlarge the mechanism by which we receive information and clarify the reception. We turn into a larger, more sensitive radio.

In 2000, Sara Lazar, a neuroscientist at Massachusetts General Hospital and a expert in functional magnetic resonance imaging (fMRI), confirmed that this process produces actual physical changes.

Conventional MRI employs radio-frequency waves and a powerful magnetic field to view the soft tissues of the body, including the brain. 'Functional' magnetic resonance imaging, on the other hand, measures the minuscule changes in the brain during critical functions. It confirms where and when stimuli and language are being processed by measuring the increase in blood flow in the fine network of arteries and veins of the brain when certain neural networks are engaged. For scientists like Lazar, the fMRI is the closest science can get to observing a brain at work in real time.

Herbert Benson had enlisted Lazar to map the brain regions that are active during simple forms of meditation. Rather than scrutinizing more monks or other meditation 'athletes' who had devoted themselves to the contemplative life, Lazar preferred to study the effect of meditation on the millions of ordinary Americans who performed

meditation for just 20–60 minutes a day. She and Benson recruited five volunteers, who had practised Kundalini meditation for at least four years.

This kind of meditation employs two different sounds to focus and still the mind while observing inhalation and exhalation of the breath. Lazar asked volunteers to alternate between intervals of meditation and control states, during which they silently ticked off a mental list of animals.

Throughout the experiment, Lazar also monitored the biological activity of her subjects – heart rate, breathing, oxygen saturation levels, levels of exhaled CO₂, and EEG levels.

Lazar discovered that, during meditation, the volunteers had a significant increase of signalling in the neural structures of the brain involved in attention: the frontal and parietal cortex, or the ‘new’ part of the brain where higher cognition takes place, and the amygdala and hypothalamus, portions of the ‘old’ brain that govern arousal and autonomic control.

This finding was another contradiction of the received wisdom that meditation is always a state of quiescence. Her results offered yet more evidence that, during certain types of meditation, the brain is engaged in a state of rapt attention.

Lazar also discovered that the signalling in certain areas of the brain and the neural activity during meditation evolved over time and increased with meditative experience. Her subjects themselves had the impression that their states of mind continued to change during each individual meditation and as they grew more experienced.[29](#)

These results suggested to Lazar that highly concentrated focus over time might enlarge certain parts of the brain. To test this, she gathered 20 long-term practitioners of Buddhist mindfulness meditation (five of whom were meditation teachers) with an average of nine years of meditation experience. Fifteen non-meditators acted as controls. Participants meditated in turn inside an ordinary MRI scanner while Lazar took detailed images of their neural structures.

Lazar discovered that those portions of the brain associated with attention, awareness of sensation, sensory stimuli and sensory processing were thicker in the meditators than in the controls. The effects of meditation definitely were ‘dose-dependent’: increases in cortical thickness were proportional to the overall amount of time the participant had spent meditating.

Lazar’s research offered some of the first evidence that meditation causes permanent alterations in brain structure. Up until the time of her experiment, this type of increase

in cortical volume had only been linked to certain repetitive mechanical practices requiring a high degree of attention, such as playing an instrument or juggling. Here was some of the first evidence that thinking certain thoughts exercises the 'attention' portion of the brain and makes it grow larger. Indeed, the cortical thickness of these regions was even more pronounced in the older participants. Ordinarily, cortical thickness deteriorates as a result of ageing. Regular meditation appears to reduce or reverse the process.

Besides increasing cognitive processing, meditation also appears to integrate emotional and cognitive processes. In the fMRI study, Lazar found evidence of activation of the limbic brain – the primitive, so-called 'instinctive' part of the brain involved with primitive emotion. Meditation appears to affect not only the brain's reasonable, analytical 'upstairs' but also the unconscious and intuitive 'downstairs'. She had discovered greater activation in the part of the brain responsible for what is usually called 'the gut instinct'. Here was physical evidence that meditation not only increases our ability to receive intuitive information, but also our conscious awareness of it. Davidson had shown increases in the 'approach' portion of the brain the part that wants to help – in his monks, who were attempting to help humanity by meditating on compassion. They had increased the 'can I help you' portion of their brains. Lazar's meditators, however, were working on mindfulness, a state of peak attention, and that part of the brain responsible for attention had grown larger. The brain's powers of observation had increased, allowing in more information, even the kind that is received intuitively.

Some people are born with a larger-than-normal antenna and better reception than usual. This appears to be the case with the psychic Ingo Swann. Swann's psychic gifts extended to remote viewing, the ability to perceive objects or events beyond normal human vision.

He had helped to develop a remote viewing programme used by the American government and was widely regarded as one of the best remote viewers in the world. Swann once had allowed the peculiar workings of his brain to be monitored and analysed by Michael Persinger, professor of psychology at Laurentian University in Canada.

Wired to an EEG machine, Swan was asked to use his skills to identify items in a distant room. At the very moment that he was able to 'see' the items remotely, his brain showed bursts of fast activity in the high beta and gamma range, similar to that of Benson's Tibetan monks.

Those bursts of activity occurred primarily over the right occipital region, the portion of the brain relating to sight. According to the results of brain-wave monitoring, Swann

had entered a super-conscious state, enabling him to receive information impossible to access during normal waking consciousness.

When examined by MRI, Swann also showed that he had an unusually large parieto-occipital right-hemisphere lobe, the portion of the brain involved with sensory and visual input.

Persinger had found a similar neural aberration in another gifted psychic called Sean Harribance.[30](#)

When monitored with EEG and single photon emission computerized tomography (SPECT) equipment during his psychi activities, Harribance evidenced an increase in firing of the right parietal lobe. Both he and Swann had been graced with a greater capacity than normal to 'see' beyond the limits of time, distance, and the five major senses.

Science has demonstrated that by thinking certain thoughts it is possible for us to alter and enlarge portions of our brains to become a larger, more powerful *receiver*. But is it also possible to develop a larger *transmitter*?

To discover some of the qualities that enhance transmission, I would have to study 'masters' of intention who were particularly gifted at transmitting. The best place to look seemed to be among talented healers.

Cancer specialist and psychologist Lawrence LeShan, who has studied how gifted healers work, discovered that they share two important practices, besides entering an altered state of consciousness: they visualize themselves as uniting with the person to be healed and imagine themselves and that person as being united with what they often describe as the absolute.[31](#)

Cooperstein's healers had also described turning off the ego and eliminating their sense of self and separateness. They had the sense of assuming the body and vantage point of the person to be healed. One healer actually felt his body changing, with shifts of patterns and distributions of energy. Although the healers did not take on the disease or pain, they sensed it once they had visualized themselves as being at one with the person being healed. At this point of union, the healers' perception markedly altered and their motor skills diminished.

They were suffused by an expanded sense of pure present, and grew unaware of the passage of time. They lost awareness of the boundaries of their own bodies, and even experienced an altered sense of bodily image. They felt taller, lighter – almost as though they were out of their physical being – engulfed by a sense of

unconditional love. They began to observe themselves, according to one healer, only as 'a kind of a core that remains':

I'm aware of the process just being beyond me ... My intent is obviously with the person – my conscious control is completely side-stepped, like I'm standing, watching. Then something else takes over ... I don't think that I ever lose complete awareness that I'm sitting there.[32](#)

Other healers experienced a more profound loss of identity; to carry out their work, they had to be at one with the person they were healing: to *become* that person, complete with his or her physical and emotional history. Their own personal identity and memory receded and they entered into some space of joint consciousness, where an impersonal self carried out the actual healing.

Some of the healers took on a mystical identification with guardian spirits or guides, and the spiritual alter ego took over.

In Krippner's experience, certain personalities are more susceptible to merging identities than others: those who, according to a psychological test, possess 'thin boundaries'.

According to the Hartmann Boundary Questionnaire test, a test developed by Tufts University psychiatrist Ernest Hartmann to test a person's psychological armament, people with thick boundaries are well organized, dependable, defensive and, as Hartmann himself liked to put it, 'well armored', with a sturdy sense of self that remains locked around them like a chain-link fence. People with 'thin' boundaries tend to be open, unguarded and undefended.[33](#)

Sensitive, vulnerable and creative, they tend to get involved quickly in relationships, experience altered states, and easily flit between fantasy and reality. Sometimes, they are not sure which state they are in.[34](#)

They do not repress uncomfortable thoughts or separate feelings from thoughts. They tend to be more comfortable than thick-boundaried people with the use of intention to control or change things around them. In a study by Marilyn Schlitz of musicians and artists, for instance, creative individuals with thin boundaries also scored best in remote influence.[35](#)

Krippner demonstrated the relationship between thin boundaries and intention with students at Ramtha's School of Enlightenment in Yelm, Washington. Many of the techniques taught at the school – for example, focusing on a desired outcome and excluding all external stimuli, blindfolding students and having them find their way around a labyrinth – were designed to help students release their usual boundaries.

The school encouraged students to engage in imaginative fantasy, claiming that it opened untapped areas in the brain.[36](#)

Krippner and several colleagues performed psychological tests on six of the long-time students who claimed to have developed keen skills in manifesting intention.

Ian Wickramasekera, a psychologist who participated in some of the Yelm research, had developed a battery of psychological tests based on his High-Risk Model of Threat Perception.[37](#)

Wickramasekera claimed the tests identify people most likely to have psychic experiences or to be susceptible to hypnosis. Although the test was originally developed to pinpoint people at high risk of psychological problems during times of major life changes, Krippner believed Wickramasekera's model could also be used to evaluate mediums and healers. Krippner and his associates found they could readily use the test to identify people whose inflexible sense of reality blocked them from perceiving or acknowledging intuitive information. Wickramasekera's model predicted that individuals would best perform healing if they were able to block the sense of a threat when they let go of their separatist notions of self.

According to their scores, the Ramtha students had extraordinarily thin boundaries. Hartmann's own mean score, derived from tests on 866 individuals, was The Ramtha students scored 343. The only other groups Hartmann had identified with boundaries this thin were music students and people suffering from frequent nightmares.

The Ramtha students also showed a high degree of what psychologists call a type of 'dissociation' (the ability to undergo profound disruptions in their attention) and a high degree of absorption (a tendency to lose themselves in ongoing activity such as hypnosis and a readiness to accept other aspects of reality).[38](#)

In my own examination of healers, I had come across two types. Some regarded themselves as the water (the source of healing); others saw themselves as the hose (the channel for healing energy to travel through). The first group believed the power resulted from their own gift. By far the largest group, however, comprised the channellers – those who acted as vehicles for a greater force beyond themselves.

Elisabeth Targ's AIDS project had recruited 40 healers of every persuasion.[39](#) Approximately 15 per cent were traditional Christian religious healers, who used the rosary or prayer.

Others were members of non-traditional healing schools, such as the Barbara Brennan School of Healing Light, or those taught by Joyce Goodrich or Lawrence

LeShan. Some worked on modifying complex energy fields through changing colours or vibrations or the patient's energy field.

More than half the healers concentrated on healing a patient's chakras, or energy centres of the body; others worked with tones, reattuning their patients with audible vibrations.

A *Qigong* master from China sent harmonizing *Qi* to the patients.

One man working in the Native American tradition went into a trance during a traditional drumming and chanting pipe ceremony on the deserted ridges of Chaco Canyon, New Mexico, and claimed to have contacted spirits on behalf of the patients.

Much of the imagery the healers used to describe what they did was framed in terms of relaxing, releasing or allowing spirit, light or love in. For some healers, the spirit was Jesus; for others, Starwoman, a healing Native American grandmother image.

Targ had interviewed the healers about their work, and I spoke with her before she died about the common threads she had discovered among their diverse approaches.[40](#)

She found that a quality of loving compassion or kindness was essential in sending out a positive intention to heal.

But no matter what their approach, most of them agreed on a single point: *the need to get out of the way*. They surrendered to a healing force.

They had framed their intention essentially as a request – *please may this person be healed* – and then stepped back.

When Targ examined those patients whose illness had most improved, and analysed which healers they had been exposed to, those healers who were the most successful were the 'channellers' – the ones who had moved aside to allow the greater force in. None of the healers who had been successful believed he possessed the power himself.[41](#) Psychiatrist Daniel Benor, who has accumulated and catalogued virtually every study of healing in four volumes[42](#) as well as on a website,[43](#) has examined the statements and writings of the most famous healers describing how they work.

One of the most remarkable and best-studied healers, Harry Edwards, wrote that a healer worked by handing over his will and his request for healing to a greater power:

This change may be described (inadequately) as the healer feeling a sense or condition enshrouding him, as if a blind had been drawn over his normal alert mind.

In its place he experiences the presence of a new personality – one with an entirely new character – which imbues him with a super-feeling of confidence and power.

...

[While engaged in his healing] the healer may be only dimly aware of normal movement, speech, etc., taking place around him. If a question is addressed to him about the patients' condition, he will find himself able to respond with extraordinary ease and without mental effort – in other words, the more knowledgeable personality of the Guide provides the answer.

Thus does the healer 'tune-in' – it is the subjection of his physical sense to the spirit part of himself, the latter becoming for the time being the superior self under the control of the director.[44](#)

To Edwards, the most important act was *moving aside*, shedding the personal ego, *making a conscious attempt to get out of the way*.

Cooperstein's healers described their experience as a sense of total surrender to a higher being or even to the process. All believed that they were a part of a larger whole.

To gain access to the cosmic, non-local entity of true consciousness, they had to set aside the limiting boundaries of the self and personal identity, and merge with the higher entity.

With this change of consciousness and expanded awareness, the healers felt they got onto an open line to this larger information field, which offered them flashes of information, symbols and images.

Words would appear, seemingly from nowhere, giving them a diagnosis. Something beyond their conscious thought would carry out the healing for them.

Although the lead-up to healing was accomplished through consciously directed thought, the actual healing often was not. In giving a 2-minute treatment, for instance, they might have a minute and a half of rational thought and then 'a five-second thing that would be an irrational thing, a space that may be the apex, the key to the whole experience'.[45](#)

The most important aspect of the healers' process was undoubtedly their surrender – their willingness to give up their sense of cognitive control of the process and allow themselves to become pure energy.

But was this capacity to *move aside* important in all types of intention? I found an interesting answer in a study of people with brain damage. Investigators at the Behavioural Neurology Program and Rotman Research Institute at the University of Toronto attempted to replicate the work of the Princeton PEAR lab using random event generators, but with one important twist: they had enlisted several patients with frontal-lobe damage. The patients who had suffered right frontal-lobe damage, which probably affected their ability to focus and maintain attention, had no effect on the machines.

The only person to have a greater than normal effect was a volunteer with a damaged left frontal lobe but whose right frontal lobe was intact. The investigators speculated that the volunteer's particular handicap could have given him a reduced sense of self, but a normal state of attention. Achieving a state of a reduced self-awareness – a difficult state for an ordinary person to achieve – might allow for greater effects of intention on the machines.[46](#)

Krippner suspects that during some altered states of consciousness, the body naturally 'switches off' certain neural connections, including an area near the back of the brain that constantly calculates a person's spatial orientation, the sense of where one's body ends and the external world begins. During a transpersonal or transcendent experience, when this region becomes inactive, the boundary in the relationship between the self and the other blurs; you no longer know where you end and someone else begins.

Eugene d'Aquili, of the University of Pennsylvania, and Andrew Newberg, medical doctor at the university hospital's nuclear medicine programme, demonstrated this in a study of Tibetan monks. Moments of meditative experience showed up as more activity in the brain's frontal lobes with less activity in the parietal lobes.[47](#)

Meditation and other altered states can also affect the temporal lobes, which house the amygdala, a cluster of cells responsible for the sense of 'I' and our emotional response to the world: whether we like or dislike what we perceive.

Stimulation of the temporal lobes or disorder in them may create familiarity or strangeness – common features of a transcendent experience. Intense focus with intention on some other being appears to 'switch off' the amygdala and so remove the neural sense of self.

Davidson, Krippner and Lazar demonstrated that we can remodel particular portions of our own brains, depending on our different types of focus and indeed different thoughts. It became clear to me that the intense focus of certain types of meditation

can be a portal to hyperspace and peak awareness, transporting the meditator to a different layer of reality.

It can also be an energizing practice more than a calming one, that can help us rewire our brains to improve our *reception* and *transmission* of intention.

I had assumed that intention was like a strong ‘oomph’, or mental push, through which you project your thoughts to another person to ensure that your wishes are carried out. But the healers described a very different process: intention requires initial focus, but then a type of surrender, a letting go of the self as well as of the outcome.

Notes Chapter 5: Entering Hyperspace

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CHAPTER SIX

In the Mood

MITCH KRUCOFF WAS RETURNING HOME from India in 1994 with alm every idea he had held about the practice of medicine turned on its head. Krucoff, a cardiologist at Duke University Medical Center, and his nurse practitioner, Suzanne Crater, had been invited to inspect the Sri Sathya Sai Institute of Higher Medicine, hospital in Puttaparthi, at the end of its first year of operation.

The hospital was the pet project of the Indian guru Sri Sathya Sai Baba, who wanted to make available the services of a modern Western hospital to the poor and needy, entirely free of charge. Krucoff had been recruited as its cardiac specialist, to advise on the technology needed to build a state-of-the-art facility for high-tech cardiac catheterizations.

Krucoff and Crater were astonished by what they had seen. The overwhelmingly spiritual dimension of the facility – even the special quality of the sound and light – had dwarfed its considerable technological achievements.

Spirituality was present in the very design of the building – in the Hindu images lovingly chosen to grace the walls. Situated 9 kilometres from Sai Baba's ashram, the building resembled an elongated Taj Mahal.

The wings had been structured as a curvature, like a welcome embrace for all those approaching its doors, and the rotunda inside the entrance was meant to represent a heart whose apex was pointing to heaven.

During their rounds, Krucoff and Crater had been struck by the effect this had on the patients – many of them Indians from extremely remote areas who had never seen running water before.

Despite the fact that they had been diagnosed with a life-threatening illness and were set to face an imposing twenty-first-century digital cath lab, not one of them seemed the slightest bit afraid. This utter absence of fear contrasted starkly with the terror and despair to which Krucoff had grown accustomed among the cardiac patients he regularly saw back home.

Krucoff longed to introduce some of these practices to hospitals in America, but if he were going to convince any of his colleagues in cardiology, he would have to prove the benefit of spirituality to the practice of heart surgery through hard data showing a measurable physiological effect. He would have to demonstrate that intangible aspects like intention, or spiritual beliefs, or even a spiritual, uplifting environment, could really make a difference to a patient's outcome.

During the 18-hour flight home, Krucoff and Crater began teasing out ideas for a study. The only way to do it, they eventually realized, was to put prayer to the test – the biggest test of its kind.[1](#)

When Krucoff got home, he began researching the scientific literature for any evidence that prayer had improved medical outcomes. Fourteen well-conducted trials of prayer had shown a positive effect. In the most famous, published by Randolph Byrd in 1988, a group of born-again Christians outside a hospital had prayed for patients in a coronary care unit. Those who had been prayed for had significantly fewer symptoms, and needed fewer drugs and less medical intervention.[2](#)

A Mid-America Heart Institute study, published around the time Targ published her AIDS study and considered at the time to have bolstered Targ's findings, showed that Christians of all denominations enlisted to pray for hospitalized cardiac patients reduced symptoms by 10 per cent, with fewer medical setbacks.[3](#)

Prayer is viewed as a kind of super-intention, a joint endeavour: you do the intending, and God carries it out. In some quarters, intention is considered synonymous with prayer, and prayer synonymous with healing; when you send out an intention, God puts the intention into action.

Indeed, many consciousness investigators consider these early prayer studies intention experiments. The small studies that had made use of groups of Christians to send intercessory prayers to heart patients are often construed as a group intention – an attempt by a collection of people to influence the same thing at the same time.

However promising the results of these early studies, Krucoff realized that a large-scale trial with tightened protocols was needed, and he mounted his own small pilot study. He enlisted 150 cardiac patients, recruited from nearby Durham Veterans Affairs Medical Center, who had been scheduled for angioplasty and stents.

Besides prayer, Krucoff wanted to see whether 'noetic' therapies, involving some form of remote or mind-body influence, could affect patient outcomes. He divided the patient population into five groups. In addition to standard medical treatment, four of the five were to receive one of the noetic treatments – stress relaxation, healing touch, guided imagery or intercessory prayer.

The fifth group would be given no additional intervention besides orthodox medical care. Every patient would undergo continuous monitoring of brain waves, heart rate and blood pressure, to gauge the moment-by-moment effect of these intangible healing influences.

Krucoff decided to turn up the volume on prayer to full blast. To recruit prayer groups, his nurse-practitioner assistant Suzanne Crater launched a worldwide campaign of solicitation. She wrote to Buddhist monasteries in Nepal and France and to VirtualJerusalem.com, which arranged for prayers to be placed in the city's Wailing Wall.

She phoned Carmelite nuns in Baltimore to ask for prayers during evening vespers. By the time she finished her campaign, she had enlisted prayer groups from seven denominations, including Fundamentalists, Moravians, Jews Buddhists, Catholics, Baptists and members of the Unity Church.

Each prayer group was assigned a group of patients, who were identified only by name, age and type of illness. Although Crater and Krucoff left the design of individual prayers to the groups themselves, they stipulated that the patients had to be prayed for by name and that the prayers on behalf of these patients had to concern their healing and recovery.

The prayer portion of the study would be blinded, so that neither patients nor staff knew who was going to be prayed for. The other noetic therapies would be administered an hour after the patients had undergone the angioplasty.

The results were impressive. Patients in all the noetic treatment groups enjoyed 30–50 per cent improvements in health during their hospital stay, with fewer complications and a lower incidence of narrowing of the arteries compared with the controls.

They also had a 25–30 per cent reduction in adverse outcomes: death, heart attack, or heart failure, a worsening of the state of their arteries or a need for a repeat angioplasty. But of all the alternative therapies employed, prayer had the most profound effect.

The study was too small to yield any definitive conclusions; after all, only 30 patients had been in the prayer group. Nevertheless, Krucoff's results seemed highly promising. Krucoff and Crater, who had christened their study MANTRA (Monito and Actualization of Noetic TRAINings), published it and presented their findings before the American Heart Association.[4](#)

Even the most conservative of cardiologists were beginning to take home the message that remote healing might actually work after all, and that prayer in particular was good for the heart.[5](#)

Krucoff understood that, for his results to be meaningful, the study needed to be replicated on a far larger scale. He rolled out his study and created MANTRA II by launching into an ambitious recruitment programme, eventually enlisting 750 patients from Duke's Medical Center and nine other hospitals across America, and soliciting 12 prayer groups made up of an even larger, more ecumenical collection of the world's major religions. Christians were recruited from Great Britain, Buddhists from Nepal, Muslims from America, Jews from Israel.

Emboldened by his early success, Krucoff and Duke loudly trumpeted the project as the largest multicentre study of remote influence, the supreme test of prayer.

With MANTRA II, he divided the patients into four groups. One group would receive prayer; another, a specially designed programme that included music, imagery and touch (or MIT therapy); the third group, MIT plus prayer; and the final control group, standard medical care. Immediately prior to undergoing angioplasty, those assigned to receive MIT would be instructed in a method of relaxed breathing while visualizing a favourite place and listening to calming music of their choice. They would then receive healing touch for 15 minutes from a trained practitioner. These patients could also wear headphones during surgery.

The point of the new study was to examine whether prayer or the noetic interventions would prevent further cardiovascular events in the hospital, such as death, new heart attacks, a need for additional surgery, readmission to the hospital, and signs of a sharp rise in the enzyme creatine phosphokinase, an indication that the heart has suffered damage. This time, Krucoff also wished to investigate longer-term effects as 'secondary endpoints': whether the interventions could alleviate emotional distress, or prevent death or rehospitalization at any point six months after the patients had been discharged.

Krucoff's study fell right in the midst of the terrorist attacks of 9/11 and their aftermath. For three months, patient enrolment in the study fell so sharply that he had to amend its design. He developed a 'two-tier' prayer strategy by recruiting 12 'second-tier' prayer groups. As soon as new patients were added to the study, the second-tier groups were to pray for the prayers of the 'first-tier' prayer groups, who had been praying for the patients all along. Through this strategy Krucoff hoped that newly enrolled patients would receive a higher 'dosage' of prayer to approximate the amount received by his patients enlisted earlier in the study.

After the enormous advance publicity, Krucoff's findings were an enormous letdown. When the results were finally in and tallied, there was no denying it: there were no differences in outcomes between any of the various groups during their hospital stay. The only apparent benefit was a slight reduction in distress among the MIT patients prior to the surgery. Otherwise, the large-scale MANTRA was an utter failure. Prayer did not seem to make anybody better.[6](#)

Among the long-term effects, there had been some therapeutic effects in alleviating emotional distress, need for further hospitalization, and even death rates after six months, but these were not considered statistically significant and they hadn't been the main focus of the study.

Wresting a small victory from this enormous defeat, Krucoff managed to get his findings published in the prestigious British medical journal, *The Lancet*. To the public, he maintained that he was 'thrilled' with the findings and that they had been misinterpreted. Krucoff's study appeared to vindicate the sceptics of prayer as a subject for scientific inquiry. The simple message appeared to be that getting someone to pray for you just does not work.

Meanwhile, in 1997, the Mayo Clinic had begun a two-year study of patients with cardiovascular disease who had been recently discharged from its coronary care unit. Nearly 800 patients were subdivided into two groups: high-risk (those who had one or more risk factors, such as diabetes, a prior heart attack or pre-existing vascular disease) and low-risk (those who had no risk factors other than their present symptoms). The two groups were again divided into two.

In addition to ordinary medical treatment, one group in each of the two categories was to receive the prayers of five people once a week for 26 weeks. The two other groups would simply continue with standard medical treatment.

At the end of the study, the investigators concluded that prayer made no difference in mortality, future heart attacks, need for further intervention or hospitalization. Although there were small differences between the treated and untreated groups, particularly among the low-risk patients, these results were not deemed to be significant.[7](#)

To settle the matter once and for all, Herbert Benson came forward with an ambitious plan. Benson had managed to straddle both mainstream and complementary camps in medicine and was well respected for it – a diplomat with the status of elder statesman between two suspicious factions. Besides his Harvard Medical School credentials, he had set up the Mind/Body Medical Institute, which was devoted to the study and practice of mind–body healing techniques. He had even coined the term ‘the relaxation response’ to describe their effects.[8](#) Lending his name to a study of prayer would legitimize it among the conservative camps.

For this study, Benson recruited five other powerhouses of medicine in the USA, including the Mayo Clinic. His plan was that this study of prayer, which he had dubbed STEP (Study of Therapeutic Effects of Intercessory Prayer), would be the largest, most scientifically rigorous of all time.

The study recruited 1800 patients undergoing coronary artery bypass surgery and divided them into three groups: the first two groups were uncertain whether they were going to receive prayer or not; the first group received prayer and the second did not. The third group, which would definitely receive prayer, was also told of the fact. Benson settled on this particular design so that he could isolate two potential effects: whether being prayed for in itself worked, and whether knowing you were going to be prayed for had any additional benefit. In this way he could control for the effect of belief.[9](#)

For his prayer groups, Benson enlisted a group of Roman Catholic monks and members of three other Christian denominations: St Paul’s Monastery in St Paul Missouri; the community of Teresian Carmelites in Worcester, Massachusetts, and Silent Unity, a Missouri Unity prayer ministry outside Kansas City. He maintained that his prayer groups included no members of Islam or Judaism because he could not find non-Christian groups happy to work within the demands of the study schedule. The prayer groups were given the patients’ first names and the initials of their surnames. Although the design of their prayers could be individual, they had to include the phrase: ‘for a successful surgery with a quick, healthy recovery and no complications’. The groups were then followed for 30 days and any post-operative complications, major events or deaths tracked among all groups.

The results shocked the world and bewildered the researchers, most of all Benson, who had spent much of his career promoting the beneficial effects of the mind on the body. The researchers had predicted the greatest benefit in the prayed- for-and- knew-it group, the second greatest effect in the prayed-for-but-didn’t-know-it group and the least effect among the didn’t-get-prayed-for-and-didn’t-know-it group.

But their results indicated that no amount of prayer under any condition, whether the patients knew it or not, made any difference to the outcome of their operations. Indeed, the results were the very opposite of the researchers' expectations. Those patients who were prayed for and knew they were being prayed for were worse off, by a statistically significant degree: 59 per cent of the prayed-for-and-knew-it group suffered post-operative complications, compared with 52 per cent among the non-prayed-fors.

Even the prayed-for-but-didn't-know-it group suffered slightly more heart attacks and strokes than those who had not been given prayer. Among the uninformed patients who had received prayers, 10 per cent suffered major complications of the surgery, compared with 13 per cent of those who did not receive prayer.[10](#)

Benson and his co-authors didn't know what to make of these results. They even wondered if the patients had suffered from a type of 'performance anxiety' as a result of the undue pressure and expectations created by the prayers.

Many commentators concluded that this study proved that prayer not only does not work, it is bad for you – or at least it cannot be scientifically tested. Krucoff, who was asked to write a commentary about the study, emphasized that prayer indeed had an effect – a negative one. People needed to discard the universally held view that being prayed for is 'a priori' good for you as these results impelled one to consider that not simply 'voodoo and spells' but also 'well-intentioned, loving, heartfelt healing prayer might inadvertently harm or kill vulnerable patients in certain circumstances'.[11](#)

The *American Heart Journal* released the study online, and its authors held press conferences. Benson cautioned the media that STEP was not the last word on prayer, although it did raise questions about whether patients should be told about prayers being offered for them.

A patient's awareness of being prayed for was considered the most important subject about prayer for future study. But others were not sure whether prayer should or could be studied any more. The John Templeton Foundation had spent \$2.4 million on the study, and with negative results like these it was likely that theirs would be the last funds available.

The STEP findings seemed to undercut my own plans for a large intention experiment. Then as I mulled over the negative findings, I came to think that the very designs of the studies might have been responsible. Although the studies attempted to be rigorous, in many instances they violated the most basic rules of scientific research.

For instance, all of the failed studies did not clearly formulate the content of the healing intention, and left the content of the prayers up to the individual supplicant. Although Benson asked that the single phrase ‘for a successful surgery with a quick, healthy recovery and no complications’ be included, he had not asked them to be specific.

The most successful intention experiments incorporate a highly specific target into the intention. In Targ’s study, the healers were given the immune system T- cell counts of the AIDS patients and they sent healing specifically to improve the counts.

The prayer groups should have been instructed to ask for a specific outcome in cardiac symptoms, or fewer cardiac stents placed during the study time, or any other highly specific request, rather than a nebulous, highly generalized statement about the patient improving.

None of the studies tightly controlled for the number of people involved in the prayer groups or for either the frequency or length of time they were to pray, which again might have confused the mass intention. Perhaps, since they were using highly diverse prayer groups, their prayers were not equivalent. In Benson’s study, the prayer groups were allowed to pray anywhere from 30 seconds to several hours four times a week.

His researchers never recorded how long the individuals prayed. In Targ’s study, although diverse healers were used, they rotated patients, so that each received only a single healing message at any one time.

As Bob Barth, director of the Office of Prayer Research, put it: ‘How do yo determine a dose of something as intrinsic as prayer? For example, is one 5-minute prayer by a Buddhist different from 10 Catholic nuns in prayer for an hour or more? Is prayer more effective once or 20 times a day?’

In commenting on Krucoff ’s findings, *The Lancet* also aired its reservations about his study design. ‘Could a more restricted denominational approach have influenced the outcome?’[12](#)

Benson’s attempt to standardize the prayer methods used in his study inadvertently interfered with the methods by which the prayer groups usually carry out intercessory prayer.

In ordinary circumstances, when prayer groups are asked to pray for someone, they request specific details about the patient, including full name, age, medical condition and periodic reports of the patient’s progress. Often they meet with the patient and

his or her family. By gathering this personal information, they are able to personalize the prayers.

Benson's study design allowed for the prayer groups only to be given the name and a last initial of the person to be prayed for. The limited information made it impossible for the prayer groups to establish a meaningful connection with or indeed even to zero in on the people they were praying for – one of the conditions that Schlitz and Radin consider important for effective remote influence.

Several groups in Benson's study objected to the design of the study. As one commentator wrote, 'This would be similar to the concept of attempting to make a cell phone call to a friend and expecting her to answer when you have only dialled the first three digits of the phone number.'[13](#)

Like STEP, Krucoff's studies did not reveal anything about the patients in order to create a connection. In Targ's research, the healers had been given a photo and a name as well as information about the patient's condition. None of the groups tested the difference between praying for a patient whose full details were disclosed and simply praying for someone with a first name and last initial.

The selection of the prayer groups was equally unscientific. None of the major prayer studies used any criteria to select participants in the prayer groups or kept track of their size or experience in prayer. Targ had selected only those healers who were highly experienced and committed with a long track record of successfully healing. Although Schlitz's Love Study employed amateurs sending healing intention, training was provided to ensure a homogeneous approach.

Another problem was the lack of a genuine control group in any of the studies. To be truly scientific, a study must be 'randomized' and randomly select participants in one group that is given the treatment and compare its outcome with a group not exposed to the treatment.

However, in any health crisis, family members routinely turn to prayer. The odds were overwhelming in all the major prayer studies that the not-prayed-for people were being prayed for by their own loved ones. In MANTRA II, 89 per cent of the patients from both treatment and control groups admitted that someone in their family was praying for them. These patients lived in the religiously active American Bible Belt.

The lack of a pure control group ultimately muddies the results of a study. This problem occurred with the early studies investigating the potential of hormone replacement therapy (HRT) to cause cancer.

Many such studies were tainted because it is virtually impossible to enlist women for study who have not taken some form of exogenous hormones – the birth-control pill, the morning after pill or HRT – at some point in their lives. Consequently, none of the studies has a clean control group of true ‘non-takers’, with which to compare results. Women who take hormones now are compared with women who have taken hormones in the past. Both situations carry a cancer risk. The same ‘tainting’ would apply to these prayer studies. People in the ‘treatment’ groups getting prayed for are being compared with patients whose relatives are praying for them.

The large prayer studies had other basic flaws. In both the Benson and Krucoff studies, the people praying did not know the patients and so would not have had a strong motivation to heal, as the ‘senders’ had in the Love Study. In Benson’s study, as Krucoff pointed out in his commentary about STEP, there should have been a true placebo group, which would have no expectation of the possibility of prayer and also there should have been a comparison between such a group and a super-group, whose members included all those exposed to prayer.

No analysis compared the effect of being prayed for with the particular belief a patient held about which groups he or she had been assigned, which would have shed light on the possible role of a placebo effect. The researchers also had not taken into account any possible stress on the patient from having to hide his or her assignment in the study from the hospital staff.[14](#)

Like STEP, Krucoff’s study violated the basic rules of scientific design, largely because of events beyond his control. When he reconstituted his study in the wake of 9/11, some of the patients received straightforward prayer from diverse prayer groups, and the others, who had been enrolled after the World Trade Center tragedy, received the ‘two-tier’ type of prayer, in which those doing the praying were themselves prayed for. Unlike the most basic of scientific trials, his study did not offer the participants the identical treatment.

Even Targ had complained about problems in study design of the very first major prayer study by Randolph Byrd, in which ordinary Christians had been asked to pray for cardiac patients. There was no information about who was taking blood pressure medication, so it was unclear whether prayer or medicine had done the healing.

There were no controls for mental attitude during the study. A high number of patients with a positive outlook may have landed in the treatment group.

Sometimes a placebo effect, an expectation of healing, can be a large factor in positive results. In one healing study of patients suffering from clinical depression, all the patients improved, even the control group, which did not receive healing, largely from the psychological boost created by the possibility of healing.[15](#)

In Benson's study the prospect of prayer might have had the opposite effect. According to Larry Dossey, the elegant Southern internist and author of many books on prayer,[16](#) the STEP study offered prayer as a 'tease', dangled in front of seriously ill patients as something they might or might not be lucky enough to get.

'Nowhere on earth is prayer delivered in this fashion,' says Dossey. 'When prayer occurs in real life, we don't taunt our loved ones with it. They are extended compassionate prayer unconditionally and without equivocation. Who can say what emotions – resentment? hostility? – were generated in these three groups of patients as a result of how prayer was offered?'[17](#)

The fact that the people who knew they were being prayed for not only had no placebo response but also evidenced more post-surgical complications than any other group, he says, 'suggests that very strange internal dynamics were operating within the Harvard prayer study.'[18](#)

The Mid-America Heart Institute study – the study in which prayer by Christian of diverse denominations had reduced symptoms in heart patients by 10 per cent – was also criticized for offering so many endpoints that it was bound to show a positive result.[19](#)

The negative results of these large prayer studies could be because praying for others does not work, because prayer simply cannot be subjected to scientific study, or simply because these new studies themselves were asking the wrong questions.

After all, according to Bob Barth of the Office of Prayer Research, these studies only represent a small proportion of prayer research.[20](#) Of the more than 227 studies investigated by the office, 75 per cent show a positive impact.

Nevertheless, to study the effect of remote intention, it may be best to move away from prayer, which contains a good deal of emotional baggage. Targ tried to isolate the effect of simple healing intention, which is different from prayer. With intention, the agent of change is human; with prayer it is God.

Simple healing intention can be more easily controlled for in a scientific study by ensuring that every member of the group sending the intention was sending the exact same message. For the purposes of my intention experiments, a simple intention to heal or improve something might avoid all the problems associated with studying prayer.

Unlike prayer, healing has been persuasively proven; a large body of evidence exists about the positive effects of distant healing – perhaps 150 studies in all.[21](#) These scientific studies have been subjected to overall reviews that rate both the

significance of the effects and the outcome. In the most cautious of such analysis, Professor Edzard Ernst, the exacting and skeptical chair of complementary medicine at Exeter University in Britain, concluded that of 23 studies, 57 per cent had shown a positive effect.[22](#)

Among the most rigorously scientific (those with double-blind trials), the average effect size, or size of change among those treated, was 0.40 – about 10 times better than the effect size of aspirin or propranolol, two drugs considered highly successful in preventing heart attacks.

Hidden in the failure of the large prayer studies lies vital instruction not only about the design of such mass experiments, but also about those elements that maximize the power of intention.

To be successful, an intention may require other parameters besides trained attention, getting out of the way, and formulating a simple request to the universe. As Gary Schwartz learned during his own research on healing, the attitude of the healers as well as the patients may matter a good deal.

Schwartz's research began as a simple study of healing intention by Reiki practitioners. Schwartz had enlisted his colleague, Beverly Rubik, founding director of the Center for Frontier Sciences at Temple University, Philadelphia, a biophysicist interested in subtle energies.

As Rubik was well versed in studies using bacteria, they decided to use as their subject *E. coli* bacteria, which had been severely stressed. One way to stress bacteria is to shock them with a sudden blast of heat. Schwartz, Rubik and their colleague Audrey Brooks carefully managed the amount of heat so that it was enough to stress the bacteria without killing off the entire sample.

They then asked 14 practitioners of Reiki to heal the bacteria that survived by transmitting a standard Reiki treatment for 15 minutes. Each practitioner was to heal three different samples over three days. Equipment with an automated colony counter kept track of the number of bacteria that survived.

Initially, Schwartz, Rubik and Brooks were surprised to find that the Reiki practitioners made no difference to the overall survival of the viable bacteria. On closer look, however, they discovered that the Reiki practitioners seemed to be successful on certain days, but not on others. This spotty batting average puzzled them.

Perhaps, Schwartz thought, a healer's success depended on some sort of connection with the subject. It was difficult, after all, to feel any warm and fuzzy connection with *E. coli* bacteria, which ordinarily resides peacefully in the gut but can wreak havoc

when it migrates out of the digestive tract. But what if he managed to get his practitioners in healing mode?

In the next batch of studies, Schwartz and his colleagues asked the Reiki practitioners to work for 30 minutes on a human patient suffering with pain, and then set them back to work on their bacteria samples.

This time, the healing was successful; the scientists discovered significantly more bacteria in the healed samples than in the controls. The healers appeared to enjoy a higher success rate once their healing 'pumps' had been primed.[23](#)

Nevertheless, Schwartz and the other researchers continued to discover instances in which the healers had a deleterious effect on the bacteria. It occurred to them that a healer's own well-being might affect results. They needed a simple test to assess true well-being, to gauge more than physical condition.

They decided to use the Arizona Integrative Outcomes Scale (AIOS), an ingeniously simple visual mean of assessing spiritual, social, mental, emotional and physical well-being during the past 24 hours.[24](#)

Developed by physician and psychologist Iris Bell, one of Schwartz's colleagues at the University of Arizona, AIOS allows patients to assess more than physical symptoms.

The subjects are told to reflect on their general sense of wellbeing, 'taking into account your physical, mental, emotional, social, and spiritual condition over the past 24 hours', then to mark a point on a horizontal line between 'worst you have ever been' on the left and 'best you have ever been' on the right that, in their view, represents their overall sense of well-being in the same time period. A number of studies demonstrated that AIOS is a useful, accurate tool for pinpointing emotional wellness and a healthy state of mind.[25](#)

In their next series of studies, Schwartz, Rubik and Brooks asked the Reiki healers to assess themselves on the AIOS scale before and after they had carried out the Reiki. With this data, the scientists discovered an important trend. On days when the healers felt really well in themselves, they had a beneficial effect on the bacteria; the counts in the bacteria given the therapy were higher than in the heat-shocked controls. On days when they did not feel so well and they scored lower on the test, they actually had a deleterious effect. Those practitioners who began the healing with diminished well-being actually killed off more bacteria than naturally died in the controls. Evidently, a practitioner's own overall health was an essential factor in his ability to heal.

Schwartz and his colleagues then tried a study using AIOS with a different type of healing, called Johrei. They recruited 236 practitioners and volunteers, and asked them to fill in the AIOS scale plus a questionnaire he had created assessing emotional state of mind before and after they administered healing.

When Schwartz and Brooks compared the AIOS tests of both the healers and the patients before and after the healing, they discovered another interesting effect. Although the patients felt better after they had received the healing, so did the healers after they had performed the healing.

Giving was as good as getting for these senders. Other research showed a similar result.[26](#) The act of healing and perhaps the healing context was itself healing. *Healing someone else also healed the healer.* [27](#)

Schwartz and his fellow researchers then carried out another study of distant Johrei healing on cardiac patients – a double-blind study so that no one but the statistician knew who was receiving healing.[28](#) The primary outcomes measured were clinical reports of pain, anxiety, depression and overall well-being.

After three days, the patients were asked if they had had a sense, feeling or belief that they had received Johrei healing. In both the treatment and control groups, certain patients strongly believed that they had received the treatment and others had a strong feeling they had been excluded.

When Schwartz and Brooks tabulated the results, a fascinating picture emerged. The best outcomes were among those who had received Johrei and believed they had received it.

The worse outcomes were those who had not received Johrei and were convinced they had not had it. The other two groups – those who had received it but did not believe it and those who had not received it but believed they had – fell somewhere in the middle.

This result tended to contradict the idea that a positive outcome is entirely down to a placebo response; those who wrongly believed they received the healing did not do as well as those who rightly believed they had received it.

Schwartz's studies uncovered something fundamental about healing: both the energy and intention of the healing itself and the patient's belief that he or she had received healing promoted the actual healing. Belief in the efficacy of the particular healing treatment was undoubtedly another factor.

In the Love Study, Schlitz and Stone had stressed the importance of a shared belief system in the success of remote influence, and Schwartz's results bear this out.

In the large prayer studies, the senders and receivers of prayer did not share the same belief system about God. Most of the patients had been prayed for by a number of groups from different religions and disparate belief systems. Even Benson's Christian study employed different Christian sects, which do not share identical beliefs. It may be uncomfortable for some groups to be prayed for by people who do not share their views about the divine.

As Marilyn Schlitz pointed out, none of the clinical trials made use of what scientists call 'ecological validity'. This means that the trials were not designed to model what happens in real life.

In the Harvard study, for example, the prayer groups were instructed to pray differently from how they would normally. None of the big prayer studies tested the effect of the kind of prayers that prayer groups believe is most likely to work.[29](#)

In these studies, says Dossey, 'what is being tested is not genuine prayer but a watered-down faux version of it'.[30](#) The contents and context of prayer were treated casually, as if it were no different than some new medication.

The Benson study also framed its intention as a 'negative' – asking that the patients heal with 'no complications' – countering the most basic folklore about prayer and affirmations, which stipulates that they should always be framed as a positive statement.

Ordinarily, says Schlitz, people have a meaningful relationship with the person they are praying for. Psychologist and mind-body researcher Jeanne Achterberg, of the Institute for Transpersonal Psychology in California, carried out a study at a Hawaiian hospital, using highly experienced distant healers, who selected as their 'patient' a person with whom they had a special connection.

Each healer was isolated from his patient, who was then placed in an MRI scanner. At random, two-minute intervals, the healers sent healing intentions to their patients, using their own traditional healing practices. Achterberg discovered significant brain activation in the same portions of the brains – mainly in the frontal lobes – of all the patients during times healing energy was being 'sent'.

When the same regime was tried out on people the healers did not know, they had no effect on the patients' brain activity. Some sort of emotional bond or empathetic connection may be crucial to the success of both prayer and healing intention.[31](#)

The large prayer studies may have failed because the researchers were looking in the wrong places for demonstration of an effect. A study of AIDS about to be published at the time of writing has also failed to find an effect.

Nevertheless, a highly significant number of people in the treatment group correctly guessed which group they were in, while the control group did not. As Schlitz concluded, 'The treatment group seemed to feel something; it just did not correlate with the clinical outcomes that were measured.'[32](#) The study may just have been asking the wrong questions.

Another important variable may be the kinds of thoughts experienced by the recipient during healing. Researchers have discovered that negative thoughts and visualization can have a powerfully negative effect on the body, as if the negativity is somehow infectious and these thoughts take physical form.

For instance, Pennsylvania researchers from the Center for Advanced Wound Care in Reading, Pennsylvania, have discovered that patients with slow-healing wounds often have negative thought patterns and behavioural or emotional wounds, such as guilt, anger and lack of self-worth.[33](#)

The same effect can occur with negative relationships. A recent study of couples showed that the stress of reliving an argument delays wound healing by at least a day. In an ingenious study by Ohio State University College of Medicine, the researcher

gathered together 42 married couples and inflicted small wounds with a tiny puncture device on one partner of each pair. During the first sessions, the partners held a conflict-free, constructive discussion and the wound healing was carefully timed.

Several months later, the researchers repeated the injury, but this time allowed the partners to raise an ongoing contentious issue, such as money or in-laws. This time, the wounds took a day longer to heal. What is more, among the more hostile couples, the wounds healed at only 60 per cent the rate of the more compatible pairs.

Examination of the fluids in the wounds found different levels of a chemical called interleukin-6 (IL6), a cytokine and key chemical in the immune system.

Among the hostile couples, the levels of interleukin-6 were too low initially and then too high immediately after an argument, suggesting that their immune systems had been overwhelmed.[34](#)

The person sending out an intention might also need to be sent good intentions. Krucoff's results as universally interpreted had overlooked one vital finding: the patients with the double-tier prayer groups who had been prayed for had fared far better in the secondary endpoints; their death and re-hospitalization rates over the six months after discharge were 30 per cent lower than the others.

Mortality over six months was lower among patients given MIT, and lowest of all among patients given MIT with prayer. These results had only been characterized as a 'suggestive trend', but may have been the entire point of the story. *Praying worked if the person doing the praying – or his prayers – also had been prayed for.* [35](#)

Healing and positive intention are simply an aspect of the constant two-way flow of communication between living things. In the person being sent intention, a shared belief in the power of the healing modality and a positive state of mind may enhance results.

Fritz Popp's research demonstrates that the degree of coherence of an organism's light emissions is linked to its overall state of health. When healers are healthy, in a positive state of mind and have engaged in a healing 'warm up', their light is more likely to shine brighter. The most effective healer of all may be the one who has been healed himself.

Notes – Chapter 6: In the Mood

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CHAPTER SEVEN

The Right Time

Persinger's basement vault was known as the Chamber of Heaven and Hell. Room COO2B, a disused sound booth, was a relic of the 1970s, its original fittings intact enormous nylon loudspeakers, deep orange flecked shag carpeting and a single item of furniture – a stained brown polyester armchair.

More than 2000 people had occupied the chair in pure darkness, a modified yellow motorcycle helmet on their heads, surrendering all control of their next half hour to the scientists behind the glass booth. Persinger, a neuroscientist, was god of room COO2B.

He had become expert in manipulating brain waves to yield up a divine experience, or, as he referred to it, 'a sensed presence'. With a few simple commands typed into a computer, he would instruct the helmet to send low-level magnetic fields coursing through the temporal lobes of his volunteers, abruptly switching sides of the brain to heighten the transcendent and occasionally terrifying nature of the experience.¹

Jesus had been sighted in the brown polyester reclining chair, as had the Virgin Mary, Muhammad, monks in hooded robes, knights in shining armour and a Native American deity, the Sky Spirit. Out-of-body experiences had been produced; near-death experiences relived. One journalist had been transported back to his life's most transcendent moment – the time he first laid eyes on his high-school girlfriend's perfect breasts.

Not all visitors found God. There had been imaginings of alien sightings and abductions, and even satanic ritual. One volunteer, overwhelmed by the sight of an enormous set of eyes and the smell of burning sulphur, attempted to pull himself

loose from the helmet and wrench off the blindfold and earplugs. As soon as the 500-pound door was pried open for him he fled, terrorized, from the room.

The nature of the experience all depended, Persinger and his assistants explained, on a physiological roll of the dice: the sensitivity of the left amygdala of the brain compared with its counterpart on the right. If the left is more sensitive, and you send magnetic waves coursing through it, you get heaven. If you are unlucky enough to be born with a more sensitive right amygdala, you get hell.[2](#)

Persinger had a singular passion: the subtle influences of geology and meteorology on human biology, particularly the electrical circuitry of the brain. A transplant from the American South, he had headed north in the 1960s to avoid the draft and a likely stint in Vietnam – a possibility he objected to on moral grounds – and he remained in Canada after receiving a professorship at Laurentian in 1971.

Forty years later, he seemed an unlikely draft dodger, with his three-piece pinstripe suits, gold-chain swag and watch fob, and clipped, offhand manner. This conservative posturing masked a bold curiosity that led him into exotic areas of inquiry – the rhythms of biological systems, the volatile energy of outer space, the nature of epilepsy, the source of mystical visions – disparate areas that eventually converged in his mind after an extraordinary epiphany. Persinger realized that living things are attuned not only to each other, but also to the earth and its constantly shifting magnetic energies. This remarkable revelation, built upon the discoveries of Franz Halberg, would convince me that careful timing to coincide with these energies might be vital for an effective intention.

In 1948, as a young medic at Harvard Medical School on a temporary visa from war-torn Austria, Franz Halberg was assigned an impossible task: to help find the cure for all disease.[3](#)

At the time, the cure was assumed to involve the cortical hormones secreted by the adrenal glands, which enable the body to adapt to the ordinary stresses of life. The search was on to find reasonable substitutes for the body's own scarce supply of steroids.

Halberg had been singled out to study mice whose adrenal glands had been removed and who were then injected with adrenaline in order to observe the effect on their circulating white blood cells called eosinophils. In ordinary circumstances, adrenaline will set off a predictable seesaw, causing more of the body's natural steroids to be secreted, which, in turn, lower the eosinophil count.

However, in animals or humans without adrenal glands, the count should remain static. But the cell count in Halberg's mice still seemed to fluctuate, even after he had

removed all trace of adrenal tissue. Later, after moving to the University of Minnesota, he carried on his studies with a near limitless supply of experimental mice, and came up with the same conclusions.

Even when he handled them less frequently, which should have caused less stress to the tiny creatures, he noticed more variation in cell count.

Halberg was mystified by this fluctuation, until he suddenly recognized a recurring pattern: the cell counts were always higher in the morning and lower at night.

The variation was rising and falling according to a predictable, 24-hour cycle. Halberg studied other biological processes, and discovered that many appear to run according to an in-built clock. All living things respond to the same 24-hour rhythm, in tandem with the earth's rotation. Halberg coined the terms 'chronobiology' – the influence of time and certain periodic cycles on biological function – and 'circadian' (circa = about; dia = day) for daily biological rhythms.

He created the Chronobiology Laboratories at the University of Minnesota and became known as the father of chronobiology. Chronobiology, as his lab began to discover, is a ready-made feature of organisms, not simply something learned or acquired – an inherent property of life.

Besides circadian rhythms, Halberg also discovered that living things keep in time to many other periodic rhythms; half-weekly, weekly, monthly and yearly cycles govern virtually every biological function.

The human pulse and blood pressure, body temperature and blood clotting, circulation of lymphocytes, hormonal cycles and other functions of the human body all appear to ebb and flow according to some basic, recurring timetable. These rhythms are not unique to humans, but are present throughout nature, and evident even in fossils of single-cell organisms that had existed millions of years ago.

Initially Halberg believed that the master switch for these biological rhythms was located in certain cells of the brain or adrenal glands. However, certain cycles carried on even when Halberg removed the brain cells in question – the adrenal glands – and even the brain itself. In his eighties, Halberg made his final breakthrough discovery: the synchronizer within every living thing is not internal but resides in the planets and in the sun.[4](#)

The sun is a furious star.

This huge ball of gases, with a surface temperature of around 6000°C, is encased by strong magnetic fields in the outer solar atmosphere – a recipe for periodic explosions, as the gases build up and magnetic fields intersect on the sun's surface.

Although the patch of space between sun and earth used to be considered an uneventful vacuum, 'space weather' is now understood to be weather so extreme, of such unimaginable turbulence, that if transferred to earth it would blow up the entire planet in an instant. Solar wind, a constant blast of electrified gas, dominates this interplanetary medium, soaring past the earth at speeds up to 2 million miles per hour. Although the earth's magnetic field usually deflects it, this gale can penetrate our magnetic field during moments of intense solar activity.

Sunspots – vortices of concentrated magnetic fields, visible to us as dark blobs on the sun's surface – begin to accumulate and then to disappear in fairly regular cycles, so that scientists can make some predictions about when the sun is likely to erupt.

A solar cycle of waxing and waning activity occurs, on average, every eleven years. As sunspots build up, so does the sun's aggressive behaviour. At unpredictable moments, it hurls solar flares, gaseous explosions with the energy of 40 billion atomic bombs, likely caused by the ripping apart and reconnection of strong magnetic fields.

Electrified bullets of high-energy protons from the nuclei of gases are picked up by the solar wind and flung towards earth at speeds of more than 5 million miles per hour, showering our atmosphere with radiation and ionization.

Periodically, the sun also releases a corona mass ejection, a ball of gas and magnetic fields of up to a billion tons, which also speed towards earth at several million miles per hour, causing extreme geomagnetic storms in space.

Scientists have long understood that earth is, in effect, a giant magnet with two poles – North and South – surrounded by a magnetic field that is constantly in flux.

This field encircles the earth like a donut in a region of space called the 'magnetosphere', and is kept in place by the solar wind, with a force of about 0.5 gauss or 50,000 nanotesla – about 1000 times weaker than that of a typical horseshoe magnet.

The geomagnetic fields (GMFs) differ in different regions and at varying times. Any changes in our solar system – the activity of the sun, the movement of the planets, the daily oscillation of the earth on its rotation – or geological changes on earth – the presence of ground water or the movement of the earth's molten inner core – can alter the strength of the earth's GMF on a daily basis.

Storms in space transfer some of the energy of the solar wind to the earth's magnetosphere, causing wild fluctuations of direction and speed in the particles in the earth's magnetic field. The National Oceanic and Atmospheric Administration (NOAA), which tracks these volatile space weather patterns, reckons that over any

given solar cycle, geomagnetic storms in space will occur about a third of the time, almost half of which are severe enough to interfere with modern technology.

Storms of this magnitude (G5, or maximum severity on the NOAA scale) can disrupt portions of the earth's electrical power, pipeline flow and high-tech communications systems, and disorient spacecraft and satellite navigation systems. In March 1989, one such storm left 6 million people in Montreal without electric power for nine hours.

At the time Halberg made his discoveries, geomagnetic storms were known to have a profound effect on the movement and orientation of animals such as pigeons and dolphins, which make use of the earth's geomagnetic field to navigate.

Biologists assumed that the earth's weak magnetic field had little effect on basic biological processes, particularly as living things have daily exposure to the more powerful electromagnetic and magnetic fields generated by modern technology. But in the course of investigating the health implications of space flight, the Soviet researchers uncovered evidence that natural geomagnetic fields, particularly those of extremely low frequencies (less than 100 hertz), have a pronounced effect on virtually all cellular and chemical processes in living things.

When Russian scientists at the Space Research Institute of the Russian Academy of Sciences explored the effects of space weather on cosmonauts being sent into space, they discovered that protein synthesis in bacteria cells is highly susceptible to changes in geomagnetic fields, and that this disturbance in protein synthesis also affects human micro-organisms.[5](#)

Geomagnetic disturbances influence the synthesis of micronutrients in plants; even single-celled algae respond to solar-cycle flux.[6](#) So attuned are plants and microorganisms to these changes that the Russian researchers made use of them as a sensitive barometer for geomagnetic disturbances.[7](#)

The Soviet scientists also discovered that if the cosmonauts suffered cardiac arrest, it was usually during a magnetic storm.[8](#) Illness on earth also appeared to parallel geomagnetic activity in space; both sickness and death increased on stormy geomagnetic days.[9](#)

But of all the systems in the body affected, changes in solar geomagnetic conditions most disturbed the rhythms of the heart.

The Space Research Institute scientists tracked the heart rate of healthy volunteers over an entire solar cycle and compared it with sunspot and other geomagnetic activity during that period. The healthiest heart rate is one with the greatest variation. In the Russian research, the most varied heart rate occurred

during times of the least amount of solar activity,[10](#) while heart rate variability (HRV) decreased during magnetic storms.

A disturbance in HRV most affects the autonomic nervous system, the system in the body that keeps it ticking over without any conscious intervention.

A low HRV increases the risk of all coronary artery disease and heart attack. During increased geomagnetic activity, the viscosity, or thickness, of the blood also increases sharply, sometimes doubling, and the bloodstream slows down.[11](#)

Sudden cardiovascular death also appears to be linked with solar geomagnetic activity.[12](#)

Heart-attack rates rise and fall according to solar-cycle activity:[13](#) the largest number of sudden deaths from heart disease occurred within a day of a geomagnetic storm.[14](#) Halberg himself discovered a 5 per cent increase in heart attacks in Minnesota during times of peak maximum solar activity.[15](#)

It is not surprising that biological systems like human beings are sensitive to external signals, such as geomagnetic disturbances. Magnetic fields are caused by the flow of electrons and atoms with charge, known as ions, and whenever magnetic forces change, they alter the direction of the flow of these atoms and particles.

Ultimately, since living organisms are also composed of particles like electrons, any profound change of magnetic direction may markedly alter their biological processes.

Once Halberg understood the effect of the earth's geomagnetic field on living things, he renamed his life's work 'chronoastrobiology' – the rhythms of biology as affected by astral bodies. The sun was the giant metronome setting the pace for all of life.

Persinger's interests had mostly to do with geomagnetic effects on the brain. Researchers in the Soviet bloc had also discovered that space weather can affect neurological processes.

Scientists at the Azerbaijan National Academy of Sciences at Baku used a special device enabling them to continuously monitor the electrical activity of the heart and brain in a small number of healthy volunteers, and to compare those rhythms with those of the earth's geomagnetic field.

They discovered that geomagnetic activity has a strong influence on brain functioning. During magnetically stormy days, EEG readings get destabilized.[16](#)

Geomagnetic turbulence also disturbs the balance between certain parts of the brain and profoundly disrupts communication within the nervous system, over-activating

certain aspects of the autonomic nervous system and lowering others.[17](#)

The sun's activity also affects mental equilibrium. As Persinger discovered, the more unsettled the weather in space, the greater the number of patients hospitalized for nervous disorders and the greater number of attempted suicides.[18](#)

Geomagnetic disturbance also seemed to correlate with increases in general psychiatric disorders.[19](#)

Even those already suffering from mental illness get more agitated during magnetically stormy days.

Persinger grew intrigued by a possible relationship between geomagnetic fluctuations in the earth and the timing of epileptic seizures after his neuroscientist colleague Todd Murphy, who had temporal-lobe epilepsy as a young child, disclosed that he often had out-of-body experiences while having a seizure.

Some data had already linked an increase in geomagnetic activity with the timing of epileptic seizures.[20](#) Could an epileptic fit result from geomagnetic disturbance? Persinger decided to study this possibility in an animal.

He injected a batch of laboratory rats with lithium pilocarpine, which causes epileptic-like seizures in the rodents, and compared the timing of the onset of seizures about an hour after the onset of laboratory-simulated increased geomagnetic activity.[21](#)

From this, Persinger inferred that, above a certain threshold of geomagnetic activity, epilepsy is more likely to be triggered. Whenever geomagnetic activity exceeded 20 nanotesla, seizures would occur more frequently.[22](#)

Persinger then discovered a relationship between sudden death – from epilepsy or cot death – and high levels of geomagnetic activity.[23](#) Sudden, seemingly inexplicable deaths might have a rational explanation after all: people with weaker constitutions are at the mercy of the sun's restless activity.

Strong geomagnetic fields also appear to affect learning profoundly – often for the better. Increased geomagnetic activity enhances memory: rats exposed to geomagnetic fields learn mazes more quickly.[24](#)

Large fluctuations in solar activity cause other subtle effects in human behaviour and performance – for instance, the ability to perform a skilled task.[25](#)

Psychologist Dean Radin once examined the effect of GMFs on bowling. He tracked the performance of experienced bowlers over a number of periods, and then compared their scores with the geomagnetic activity of the same period.

Large geomagnetic fluctuations the day before a match appeared to cause more uneven results than normal – a 41 per cent variance in the men's scores compared with the more consistent scores obtained during days of geomagnetic stability.[26](#)

Other research has demonstrated that the greater the change in the earth's geomagnetic field, the greater the number of traffic violations and industrial accidents.[27](#)

The most important determinant appeared to be large *change* in geomagnetic activity, either from turbulent to calm or the reverse.

Although periodically destabilizing, exposure to the daily ebb and flow of earth's geomagnetic activity may be essential to life here. The Solar Terrestrial Influences Laboratory at the Bulgarian Academy of Sciences in Sofia carried out biological experiments on board the Soviet Mir space station to examine what happens to cosmonauts who are deprived of contact with the earth's geomagnetic field while in space. The scientists constructed a 'geomagnetic vacuum', a six-metre stainless steel decompression press-chamber, which partially blocked out the earth's natural geomagnetic field. Seven healthy young men were sealed off in the chamber and their bodily processes analysed. After being placed in the decompression chamber, the men evidenced a number of upsets in brain-wave activity. Sleep was more restless, with fewer periods of deep sleep.[28](#)

Contact with geomagnetic fields may play a primary role in maintaining the equilibrium of the nervous system. Indeed, the earth's tiny geomagnetic fluctuations have the most profound effect on the two major engines of the body: the heart and the brain.

Persinger went on to discover other extraordinary geophysical effects on human beings.

Electromagnetic and geomagnetic phenomena resulting from the earth's shifting plates, earthquakes, or from unusually high rainfall levels – even electromagnetic 'luminosities', or lights in the sky – can all stimulate certain portions of the brain that produce hallucinations.

Between 1968 and 1971, more than 100,000 people reported observing visions of an apparition of the Virgin Mary above a church in Zeitoun, Egypt. When Persinger examined the seismic activity in the area over the same time period, he discovered an unprecedented peak in earthquake activity.[29](#)

Sometimes the electromagnetic effects were man-made. At one point he studied a Roman Catholic woman with early brain trauma who reported nightly visitations by

the Holy Spirit.

Ultimately, he discovered the source of the miracle: her disability caused her to be unduly affected by the electric alarm clock situated near her head as she slept.[30](#)

Persinger wondered whether he could reproduce these types of geomagnetic disturbances in the laboratory. His colleague Stan Koren modified and wired up a motorcycle helmet (thereafter named the 'Koren' helmet) so that it could send out very-low-frequency complex magnetic fields – about the amount that radiates from a telephone handset – in precise directions.

Participants would be fitted in the helmet, then placed in the acoustic chamber of room COO2B, which had been especially adapted to block out electromagnetic noise. Turning on the helmet would produce what Persinger referred to as 'temporal lobe transients', something possibly like a micro-seizure – tiny episodes causing alterations in neuronal firing patterns. This produced virtually the same effect on the brain as exposure to increased ambient geomagnetic activity.

Over time, Persinger began to recognize patterns. The brain waves of his participants would fall into resonance with the complex magnetic fields and remain in synchrony for up to 10 seconds after he had turned off the helmet.[31](#)

Through trial and error, he discovered that the portion of the brain most susceptible to electromagnetic and geomagnetic effects are the temporal lobes. Sending low level (1 microtesla), pulsed magnetic fields over the right cerebral hemisphere slowed brain waves to an alpha rhythm (8–13 hertz), but only on the right side.[32](#)

Our 'sense of self' and our sense of the 'other' are housed in both temporal lobes but primarily in the left hemisphere, where the language centres are located.

To function normally, both left and right temporal lobes must work in harmony. If something upsets this balance, the brain will sense another 'self' and create a hallucination.

As Persinger discovered in his experiments, stimulating the right temporal lobe portion of the brain generates the sense, presence or feeling of spiritual visions, both good and bad.

Aiming magnetic fields at the amygdala of the brain at the same time colours the experience with intense emotion, just as occurs during a spiritual experience. By first stimulating one side of the amygdala and then the other,

Persinger found that he could heighten the emotional complexion of the experience.

Volunteers wearing the Koren helmet experienced divine epiphanies, apparitions, out-of-body sensations and even a hallucination of Satan purely through temporal-lobe stimulation.

The nature of the experience largely depended on the participant's individual history: negative early life experiences tend to increase the sensitivity of the right temporal lobe, and those with a high proportion of such experiences tend to have a negative experience while wearing the helmet. A happier person, with a more sensitive left temporal lobe, is more likely to experience a sense of the divine.[33](#)

It would have been tempting for Persinger to conclude that all spiritual experience is simply geomagnetically induced hallucination, except for one unsettling fact: extrasensory perception and other psychic abilities appear to be more acute during particular types of geomagnetic activity.

When the earth is 'calm' and geomagnetic flux at an ebb, telepathic and extrasensory perceptions increase.[34](#)

Even minor environmental changes – from slight variations in the weather to solar patterns appear to have a profound effect on extrasensory perception or the ability to view things remotely.

The reverse occurs with psychokinesis – mental attempts to change physical matter. The power of intention increases when the earth's energy is agitated.[35](#)

In the 1970s, Persinger was able to test the effects of geomagnetic activity on telepathy during sleep by teaming up with noted parapsychologist Stanley Krippner, then the director of a dream laboratory at Maimonides Medical Center in New York City.

Krippner had perfected an experimental protocol to test telepathy, clairvoyance and precognition in dreams during deep sleep. Volunteers would be paired off.

While one partner slept, the other would be in a separate room and would be asked to concentrate on an image and attempt to 'transmit' the image to the dreamer, so that it would be incorporated into his dream.

Upon waking, the participants who had been sleeping would describe their dreams in great detail, to determine whether they contained anything resembling the target pictures they had been sent during their slumbers.[36](#)

Persinger and Krippner found that participants did better on certain days than on others.

When they tracked geomagnetic activity during the period of the study, they discovered that the dreamers had significantly higher accuracy in picking up the target pictures on nights when the earth's GMF activity was relatively quiet.[37](#)

Geomagnetic activity also affects precognitive dreams – those that forecast events. Dr Alan Vaughan, a well-known clairvoyant whose dreams accurately foretold the future in great detail, kept a detailed dream diary in order to compare their contents with future events.

One of Vaughan's dreams predicted the murder of then-presidential candidate Robert Kennedy two days before he was assassinated.[38](#)

An examination of the geomagnetic activity on the nights that Vaughan had dreamed 61 such premonitions showed that it was significantly quieter on the days when he had his most accurate dreams.[39](#)

During days of geomagnetic calm, spontaneous instances of telepathy or clairvoyance are more likely to occur[40](#) and remote viewing accuracy appears to improve.[41](#)

Persinger carried out his own intriguing test of ESP using a group of couples.

One member of each pair was shown an image while it was being bathed in magnetic fields, then asked to describe the memory of an experience he or she had shared with the partner that was prompted by the image. Simultaneously, in another room, the partner was shown the same images and also asked to describe a memory.

When Persinger compared the results, he discovered that the two narratives were most alike when the ambient geomagnetic activity was at its quietest. The greater the geomagnetic activity, the less the two sets of memories mirrored each other.[42](#)

Nevertheless, the two sexes appear to respond very differently to geomagnetic activity, which Persinger discovered after comparing a database of paranormal experiences with geomagnetic activity and breaking down the data by sex.

Men tended to have more premonitions on days when geomagnetic activity was high (above 20 nanotesla), whereas women reported more premonitions if the geomagnetic activity was low (below 20 nanotesla).

Men also tended to have more accurate memories with higher geomagnetic activity; women, with lower geomagnetic activity. Just as Krippner had found, the people most susceptible to extrasensory experiences were those with 'thin boundaries', particularly those who had already had paranormal encounters.[43](#)

With time, Persinger found that he could enhance powers of extrasensory perception with the artificial geomagnetic fields of the Koren helmet. The remote-viewing ability of one of his students considerably improved after he was exposed to weak horizontal magnetic fields.[44](#)

In 1998, Persinger decided to put the Koren helmet to the ultimate test. Could it interrupt the ability of one of the greatest remote viewers in the world?

He invited Ingo Swann to his basement lab. Swann, then 68, soon proved he had lost none of his extrasensory prowess; he correctly described and drew in great detail images of randomly selected photographs sealed in envelopes in another room.

Nevertheless, after Persinger bathed the photos in complex magnetic field patterns, Swann's accuracy suddenly plummeted. The most disruptive fields had different signal wave forms of varying phases.

This suggested that Swann was picking up the information in wave form and that those signals were easily interrupted by magnetic fields that could disturb their coherence.[45](#)

As Gary Schwartz had also discovered, information transmitted or received by human beings must have a strong magnetic component.

Persinger's evidence persuaded me that geomagnetic activity influences the clarity of our reception in picking up quantum information.

But do geomagnetic fields also affect the strength of our transmissions and their effect on the physical world? Research by Stanley Krippner offers a few clues.

Krippner wished to test the hypothesis that psychokinesis is likely to occur on days when the earth is 'noisy'. He and his team worked with the Brazilian sensitive Amyr Amiden, known for his extraordinary psychokinetic ability, and set about comparing the time of Amiden's psychokinetic activities with geomagnetic fluctuations in the Brasilia area, where the sessions were taking place.

Krippner's team also took readings of Amiden's pulse and blood pressure.

The team found a significant correlation between Amiden's psychic feats and the daily geomagnetic index for the entire southern hemisphere. For instance, Amiden performed the highest number of psychokinetic feats on 10 March and 15 March, which were the days that month with the greatest geomagnetic activity. He produced nothing out of the ordinary on 20 March, the geomagnetically quietest day of the month.[46](#)

Amiden's psychic abilities were preceded by both a rise in his diastolic blood pressure (the pressure of the blood as it returns to the heart) and a rise in geomagnetic 'noise'. It may be that geomagnetic activity must first cause changes in the 'heart brain' before a person can transmit information that can affect physical matter.

Interestingly, as with couples in the Love Study, Amiden's most powerful psychokinetic effects *anticipated* strong input: in his case, geomagnetic flux.

In one instance, two religious medallions suddenly materialized in the room where Amiden and the researchers were present, appearing to drop from the ceiling – an event that was followed by a sudden rise in the area's geomagnetic field. Can humans anticipate this geomagnetic noise, and, if so, do such anticipatory windows offer them more psychokinetic power than usual?

Psychologist William Braud carried out some intriguing studies of the effect of geomagnetic fields on intention by examining whether high levels of geomagnetic activity were correlated with powers of remote influence. Braud examined the effect of sending intention to human blood cells and to another person.

Like Krippner, he discovered that the success of intention was linked to a 'noisy' sun producing high geomagnetic activity.[47](#)

Besides solar activity, other environmental factors should be considered when working out the best times to send intention.

A number of scientists, including Persinger, found that certain days and certain times of day influence the success of ESP and psychokinesis.[48](#) The best results occur around 1 p.m. local sidereal time, which is time measured by our relation to the stars, not the sun.

Local sidereal time is worked out as the hour's angle of the vernal equinox, where the plane of the earth's equator would intersect with that of its orbit, if measured out in the heavens.

Psychokinetic effects also seem to be greater about every 13 days, at times when solar wind is modulated.[49](#)

It might also be worth avoiding times of low visibility and high winds, a condition which produces a high percentage of ions with electrical charges in the air. An ion forms when a molecule encounters enough energy to unleash an electron.

They are also created by rainfall, air pressure, forces emitted during a waterfall and the friction from large volumes of air moving rapidly over a land mass, as during so-called ill winds, such as El Niño or Santa Anas of southern California.

Both positive and negative ions are equivalent to a tiny pulse of static electricity, and the air that we breathe is made up of billions of these tiny charges.

Good 'clean' air contains 1500–4000 ions per cubic centimetre, and the preferred ratio should be slightly more negative than positive ions: 1.2 to 1. However, ions are highly unstable; in our industrialized, largely indoor lives, filled with electromagnetic charge from pollution and artificial sources, this ideal number is drastically diminished and the ratio disturbed, leaving all but the most robustly outdoorsy among us inhaling too low a level of ions, with a predominance of positive ions.

Living with low levels of ions is not particularly good for us – or for our ability as receivers or transmitters.

Research in California and Israel has shown that lower concentrations of either positive or negative ions will produce fewer alpha frequencies in the human brain and that sudden higher levels of either charge can produce rapid, distinctive brain-wave changes.[50](#)

Persinger's research offers a vast amount of evidence that magnetic frequency affects our ability to 'tune' in and transmit, and also affects those portions of the brain that receive the information.

Subtle shifts in the earth's geomagnetic fields most noticeably affect the heart and brain, the very systems of the body shown by the

DMILS research and Schlitz's Love Study to be the primary source of transmission. After examining Persinger's work, I began to view intention as a vast energetic relationship involving the sun, the atmosphere, and earthly and circadian rhythms.

To send intention effectively, we would have to take account of these energies.

Persinger had usefully located not only the best 'channel' for intention, but also the best time to turn it on.

Notes – Chapter 7: The Right Time

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